

Opioids and Post-COVID Labor-Force Participation*

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Abstract

At the onset of COVID-19, U.S. labor-force participation fell by about 3 percentage points and remained below pre-pandemic levels three years later. Recovery was slower in states hit harder by the pre-pandemic opioid crisis, measured by age-adjusted overdose death rates. An event study shows that a one-standard-deviation increase in pre-COVID opioid-overdose mortality is associated with an additional 1.01 percentage-point decline in labor-force participation after March 2020. The effect is stronger among individuals without a college degree. Placebo tests using non-opioid mortality provide little evidence that broader differences in health across states explain this relationship. In high-exposure states, the slower recovery was also associated with a larger increase in disability-related labor-force nonparticipation.

Keywords: Labor-Force Participation, Opioids, COVID-19, Health, Disability

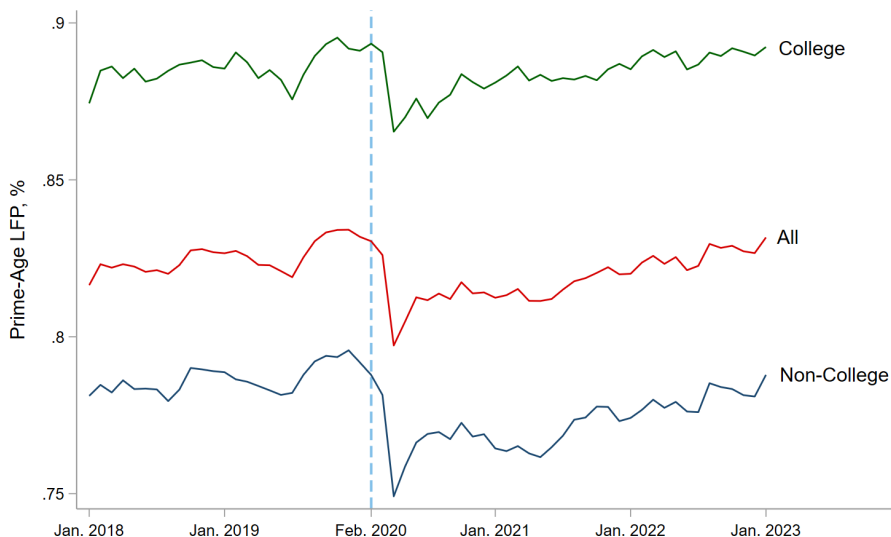
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1 Introduction

The COVID-19 pandemic caused a sharp drop in U.S. labor-force participation (LFP), followed by a slow recovery. As Figure 1 shows, labor-force participation among prime-age individuals aged 25–54 declined by about 3 percentage points. The decline was more pronounced for those without a college degree. Furthermore, the pace of recovery in labor-force participation varied substantially across U.S. states. In December 2022, the LFP among prime-age workers was still 5 percentage points below the trend in Michigan, while the LFP gap relative to the trend had disappeared in Virginia. On average, LFP was around 2 percentage points below the trend in December 2022, with a standard deviation of 2 percentage points across US states.

Figure 1: Labor-Force Participation Rate, Ages 25-54

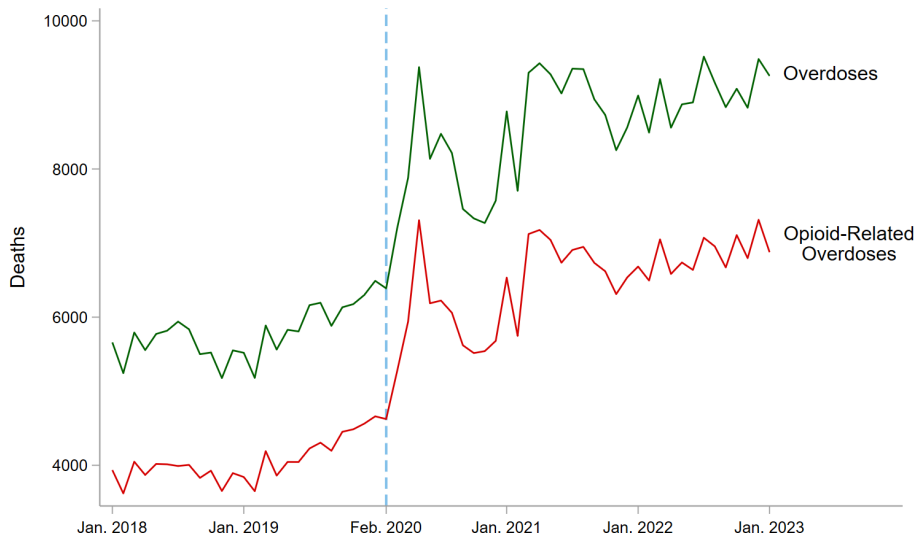


Notes: *Monthly labor-force participation rates among 25-54-year-olds by education. Source: Current Population Survey (CPS).*

The COVID-19 pandemic struck the U.S. in the midst of another unfolding epidemic: the opioid crisis (Cutler and Glaeser, 2021; Alpert et al., 2022; Greenwood, Guner, and Kopecky, 2024). Between 2000 and 2019, nearly half a million people died from an opioid overdose. Deaths from opioid overdoses increased significantly during COVID-19, as shown in Figure 2. The number of deaths rose from around 50,000 in 2019 to more than 80,000 in 2021 and 2022. Deaths from other illicit drugs, mostly methamphetamine, also increased during COVID-19. The total number of drug overdose deaths increased from about 70,000

in 2019 to over 106,000 in 2021 and 2022. The increase in drug overdose deaths during this period is also documented by [Mulligan \(2022\)](#) and [Greenwood, Guner, and Kopecky \(2022\)](#).

Figure 2: Monthly Deaths from Overdose and Opioid Overdose



Notes: *Monthly deaths from all drug overdoses and from opioid-related overdoses. The vertical dashed line marks February 2020. Source: CDC Multiple Cause of Death (CDC-MCOD).*

Individuals who misuse opioids or use other illicit drugs are significantly less likely to participate in the labor force compared to those who do not use opioids at all or use them strictly as prescribed ([Greenwood, Guner, and Kopecky, 2022](#)). We study whether pre-pandemic opioid exposure amplified the decline in labor-force participation following the onset of COVID-19. Our event-study design treats the onset of COVID-19 as a common shock and pre-COVID opioid exposure as a moderator of its effect. The central hypothesis is that, although COVID-19 increased illicit drug use nationwide, states with greater pre-existing opioid exposure experienced larger increases in illicit drug use, further slowing the recovery in labor-force participation.

Pre-COVID opioid exposure is measured using age-adjusted opioid overdose death rates from 2017, serving as a comprehensive indicator of the opioid crisis’s intensity in each state. Additionally, this measure suffers less from measurement error than other measures. Our identification strategy rests on the idea that COVID-19 led to larger increases in illicit drug use in states with greater pre-pandemic opioid exposure, generating differential changes in labor-force participation. We therefore first examine whether overdose deaths rose more in high-exposure states after the onset of COVID-19. We find that a one-standard-deviation

increase in pre-COVID opioid exposure is associated with an increase of 0.46 deaths per 100,000 residents in the monthly crude overdose death rate by May 2020 and an increase of 1.74 deaths per 100,000 residents in the annual age-adjusted overdose death rate in 2020.

The analysis then shows that a one-standard-deviation increase in pre-COVID opioid exposure is associated with an additional 1.01-percentage-point decline in labor-force participation relative to trend following the COVID-19 shock. For instance, moving from a state at the 25th percentile of opioid exposure, such as Minnesota, to one at the 75th percentile, such as Michigan or Pennsylvania, corresponds to a 1.29-percentage-point decline in labor-force participation. This is substantial, given that the standard deviation in prime-age labor-force participation throughout the post-COVID period across US states was 3.42 percentage points.

The finding is robust to multiple event-study specifications with flexible state-specific trends and longer pre-periods. The results are also robust to alternative measures of pre-COVID opioid exposure, including the age-adjusted opioid overdose death rate for the combined 2010-2017 period, the fraction of individuals with an opioid use disorder, shipments of medications for opioid use disorder, and pre-2010 OxyContin exposure. An alternative synthetic control group approach, as outlined in [Abadie and Gardeazabal \(2003\)](#), also yields results comparable to those obtained with the original approach.

We show that the finding is not driven by differences in general health across states. States with higher initial opioid exposure may have poorer overall health or weaker healthcare systems, potentially explaining the observed effects. To test this, we conduct a placebo analysis, replacing opioid exposure with age-adjusted death rates from leading non-opioid-related causes. The results show no significant differences in post-COVID labor-force participation between states with higher and lower non-opioid-related death rates.

Furthermore, we find that the effects vary by demographics. While the results are similar for both men and women, differences emerge by educational attainment and age. The effects are significant for labor-force participation among non-college-educated people, who tend to have higher rates of opioid use, but not for the college-educated. Moreover, the impact of opioids on labor-force participation is more pronounced among individuals aged 45 to 54.

Finally, we show that individuals with a drug use disorder have lower labor-force participation, poorer health, and are more likely to be out of the labor force because of disability. During COVID-19, the gap in disability-related nonparticipation between nonusers and in-

dividuals with drug use disorders widened. Moreover, states with higher pre-COVID opioid exposure experienced larger post-onset increases in the share of individuals out of the labor force because of disability.

Overall, our findings suggest that the COVID-19 shock interacted with preexisting opioid exposure in ways that amplified its labor-market consequences. The pandemic increased isolation, altered economic incentives, and disrupted access to mental-health and addiction treatment (Adams-Prassl et al., 2022a; Currie et al., 2021a; Russell et al., 2021; Mulligan, 2022). In states with greater pre-pandemic opioid exposure, these disruptions were accompanied by larger increases in overdose mortality and slower recoveries in labor-force participation. Our results therefore illustrate how one public-health crisis can interact with another to produce persistent economic effects.

Related Literature This study contributes to the growing literature on opioids and labor-market outcomes. The relationship can operate in both directions. Several studies link deteriorating economic conditions to greater opioid misuse and higher drug-overdose mortality (Hollingsworth, Ruhm, and Simon, 2017; Carpenter, McClellan, and Rees, 2017; Pierce and Schott, 2020; Venkataramani et al., 2020). Conversely, opioid access may affect labor supply through competing channels: pain relief may preserve work capacity, while dependence and other adverse effects can impair health, reduce productivity, and weaken labor-force attachment. Empirical findings differ across settings. Currie, Jin, and Schnell (2019) find modest positive effects of opioid prescribing on employment for women and no statistically significant effect for men, while Harris et al. (2020), Aliprantis, Fee, and Schweitzer (2023), Beheshti (2023), and Powell (2026) find evidence of adverse labor-market consequences of opioid exposure. Research on the transition from prescription to illicit opioids finds reductions in employment and labor-force participation (Cho et al., 2021), as well as lower hours and earnings and increases in disability applications and benefit receipt (Park and Powell, 2021). More recent work documents slower transitions from unemployment and nonparticipation into employment (Mukherjee, Sacks, and Yoo, 2025) and adverse effects of opioid initiation on job performance and retention among U.S. military personnel (Alpert, Schwab, and Ukert, 2025).

Methodologically, our approach is closely related to studies that combine a nationwide shock with geographic differences in prior opioid exposure. Alpert, Powell, and Pacula (2018) and Park and Powell (2021) exploit the 2010 reformulation of OxyContin and cross-state variation in pre-reformulation OxyContin misuse. Cho et al. (2021) use pre-2010 oxy-

codone exposure to study the broader tightening of oxycodone supply in that year, while [Beheshti \(2023\)](#) exploits the 2014 rescheduling of hydrocodone and cross-area variation in prior prescribing. We apply a related exposure-based design to COVID-19, examining whether pre-pandemic opioid exposure amplified the decline in labor-force participation following a common health shock, rather than estimating the effects of an intervention in opioid markets.

Our findings also relate to the literature on the labor-market effects of COVID-19. Early studies emphasize differences by gender and family circumstances ([Alon et al., 2020](#); [Albanesi and Kim, 2021](#)) and the role of occupational tasks, contact intensity, and the feasibility of remote work ([Bartik et al., 2020](#); [Dingel and Neiman, 2020](#); [Adams-Prassl et al., 2022b](#); [Mongey, Pilossoph, and Weinberg, 2021](#)). [Coibion, Gorodnichenko, and Weber \(2020\)](#) document the initial collapse in participation, while [Forsythe et al. \(2022\)](#) highlight the roles of temporary layoffs, recalls, and retirements in a recovery characterized by tight labor markets. Other work examines the rise in quits and job-to-job reallocation ([Gittleman, 2022](#); [Barlevy et al., 2024](#)). [Faberman, Mueller, and Şahin \(2022\)](#) document a decline in desired work hours that persisted through the end of 2021, while [Bagga et al. \(2025\)](#) show how shifts in the demand for and supply of job amenities, particularly remote work, can generate persistent labor reallocation.

More closely related to our analysis are studies of the persistence and sources of changes in labor supply after the onset of COVID-19. [Abraham and Rendell \(2023\)](#) distinguish the contributions of demographic changes and preexisting trends from the remaining shortfalls in participation and hours, and assess explanations involving infection risk, lingering health effects, and changing attitudes toward work. [Hobijn and Şahin \(2023\)](#) emphasize that estimates of “missing workers” and “missing jobs” can be overstated when preexisting participation trends are ignored or pre-pandemic cyclical improvements are extrapolated. At the individual level, [Goda and Soltas \(2023\)](#) find persistent reductions in labor-force participation following COVID-related work absences. Our analysis complements this literature by showing that pre-pandemic opioid exposure was a source of heterogeneity in the recovery: states more exposed to the opioid crisis experienced larger declines and slower recoveries in labor-force participation following the COVID-19 shock.

The rest of the paper is structured as follows: Section 2 discusses the data. Section 3 presents the empirical methodology. Section 4 shows the results from the main specification. Robustness checks are presented in Section 5, while Section 6 highlights potential mechanisms. Section 7 concludes.

2 Data and Motivating Evidence

The empirical analysis is conducted at the US state level, including the District of Columbia, covering each month from January 2018 to January 2023. The primary outcome variable is the *prime-age labor-force participation rate* (LFP), defined as the share of the civilian noninstitutional population aged 25 to 54 who are in the labor force. We compute monthly state-level LFP values from the Current Population Survey (CPS) (Flood et al., 2025).¹ CPS data on gender and educational attainment are also used to analyze outcomes of different socioeconomic groups. Respondents are classified as college-educated if they have attained at least a bachelor’s degree.

The primary measure of pre-COVID opioid exposure is the age-adjusted mortality rate from opioid overdoses in 2017, sourced from the Centers for Disease Control’s *Multiple Cause of Death* database (CDC-MCOD).² An opioid overdose death is identified when the underlying cause of death is a drug overdose, and opioids are listed among the multiple causes. For underlying causes, the following ICD-10 codes are included: X40-X44 (accidental drug poisonings), X60-X64 (intentional self-poisoning by drugs), X85 (assault by drug poisoning), and Y10-Y14 (drug poisoning of undetermined intent). Opioids as a multiple cause of death are identified using codes T40.0 (Opium), T40.1 (Heroin), T40.2 (Other opioids), T40.3 (Methadone), T40.4 (Other synthetic narcotics), and T40.6 (Other and unspecified narcotics).

Opioid-related deaths are used as a comprehensive measure of opioid exposure, encompassing both legal and illegal opioid consumption. Thus, overdose mortality captures the cumulative impact of opioid prevalence in each state. We use overdose mortality in 2017 as our baseline exposure measure to ensure that opioid exposure is predetermined with respect to the COVID-19 pandemic and measured before its onset. The results with overdose mortality for the 2010-2017 period are also presented. Other measures, such as self-reported misuse rates in surveys, are not available at a monthly frequency and can be subject to under-

¹Prime-age mortality is small relative to the prime-age population, so changes in deaths do not mechanically generate the LFP changes we study. The COVID-19 pandemic temporarily disrupted CPS data collection, particularly due to the suspension of in-person interviewing in spring 2020. While response rates declined, existing evidence suggests that the resulting bias in aggregate labor-market statistics was small and diminished as standard operations resumed (Ward and Edwards, 2021; McIllece, 2020). In our setting, nonresponse would bias the estimates only if changes in survey response were systematically correlated with pre-COVID opioid mortality across states. We are not aware of evidence supporting such a pattern.

²Available at <https://wonder.cdc.gov/mcd.html>. Causes of death are coded using ICD-10, the 10th revision of the International Classification of Diseases by the World Health Organization.

reporting, while prescription volume measures exclude illicit opioids. Summary statistics for the age-adjusted opioid overdose mortality rates across US states are presented in Table 1. Notably, the age-adjusted opioid death rate rose by over 50 percent between 2019 and 2021. Figure A1 in Appendix A.1 shows that this large increase is primarily due to a large increase in synthetic-opioid-related deaths after the onset of COVID-19. Synthetic-opioid-related deaths are mainly due to fentanyl and have been rising in the U.S. since 2013.³ There is also considerable variation in death rates across states, as shown in Figure A2 in Appendix A.1. In 2017, opioid death rates ranged from approximately 3 deaths per 100,000 people in Nebraska and Hawaii to 39 in Ohio and 50 in West Virginia. Furthermore, this disparity across states has widened since the COVID-19 pandemic, as evidenced by an increase in the standard deviation from 10.16 to 13.77 between 2019 and 2021.⁴

Table 1: Age-Adjusted Opioid Overdose Death Rate

	Mean	SD	Min.	Max.
2017	16.26	10.52	3.1	49.6
2019	16.66	10.16	3.5	43.0
2021	26.26	13.77	5.7	77.2
2022	26.82	13.03	5.6	72.5

Notes: Summary statistics for the age-adjusted opioid-related overdose death rate for 2017, 2019, 2021, and 2022. Death rates are computed per 100,000 people. Source: CDC-MCOD.

Figure 3 presents the relationship between pre-COVID opioid exposure and the post-COVID recovery in labor-force participation across U.S. states. As a measure of recovery, the gap between the observed and predicted labor-force participation for prime-age individuals during the last six months of 2022 is used.⁵ The predicted post-COVID values are based on state-level regressions of labor-force participation on a linear time trend.⁶ More than two years after the onset of COVID-19, states with higher initial opioid death rates in 2017, such as West Virginia and Ohio, had labor-force participation rates that were well below the predicted values.

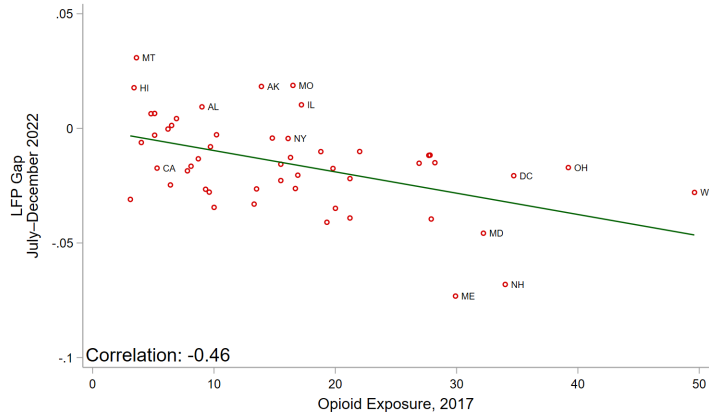
³The rise of overdose deaths involving fentanyl starting from 2013, referred to as the third wave of opioid overdose deaths, has been documented by [Ciccarone \(2019\)](#) and [Centers for Disease Control and Prevention \(2026\)](#). [Friedman and Shover \(2023\)](#) document the increasing importance of polysubstance overdose deaths involving fentanyl and stimulants.

⁴Further details on the demographic and geographic variation in fatal drug overdoses over time can be found in [Monnat \(2022\)](#).

⁵This period was chosen to capture differences across states in LFP recovery, away from the immediate impact of the COVID-19 shock.

⁶Further details on the estimation of trends are provided in Appendix A.2.

Figure 3: Relationship Between Opioid Exposure and Prime-Age LFP Gap



Notes: *Correlation between the gap in prime-age LFP in July-December 2022 and the age-adjusted opioid-related overdose death rate for 2017. Source: CDC-MCOD and CPS.*

For the analysis we also consider the age-adjusted opioid death rate for the combined period 2010-2017 and three alternative measures of pre-COVID opioid exposure. The first measure is the percentage of individuals in a state with opioid use disorder in 2017-2018, derived from the National Survey on Drug Use and Health (NSDUH). This annual survey provides national and state-level data on tobacco, alcohol, illicit drug use (including non-medical prescription drug use), and mental health. Opioid use encompasses prescription pain relievers and heroin, assessed based on usage within the past 12 months. Prescription misuse includes any use not directed by a doctor, such as use without a prescription or in greater amounts or frequency than prescribed. All heroin users are classified as misusers. Misusers are further screened for opioid use disorder, identified by health issues, disabilities, or significant life disruptions caused by recurring use. The NSDUH is also used to identify individuals who suffer from any drug use disorder, including cocaine, hallucinogens, heroin, inhalants, methamphetamine, and psychotherapeutics (pain relievers, tranquilizers, sedatives, stimulants). This disorder measure captures sustained and clinically defined dependence and abuse, and, therefore, reflects more intensive usage than simple indicators of past-year drug use.⁷

The second measure of opioid exposure is the total shipment of medications used to

⁷State-level data on drug and opioid use disorder are obtained from the Restricted-Use Data Analysis System (RDAS) of the Substance Abuse and Mental Health Services Administration, as the NSDUH public-use files lack state identifiers. These state-level estimates pool multiple survey years. For opioid use disorder, we use the variable UDPYOPI, which measures past-year opioid abuse or dependence under DSM-IV criteria.

treat opioid addiction in each state, primarily methadone and buprenorphine. Data are sourced from the Drug Enforcement Administration’s (DEA) Automation of Reports and Consolidated Orders System (ARCOS).⁸ The DEA provides the quantity (in grams) of these drugs distributed to each 3-digit zip code area, which is then aggregated at the state level. Morphine milligram equivalents (MME) are used to calculate a per capita MME amount for each state in 2017.⁹

The last measure uses cross-state variation in pre-2010 exposure to OxyContin misuse. This measure, developed by [Alpert, Powell, and Pacula \(2018\)](#), captures the rate of nonmedical OxyContin use in each state using NSDUH waves from 2004–2005 through 2008–2009, prior to the 2010 reformulation of OxyContin into an abuse-deterrent version. By focusing on pre-2010 nonmedical OxyContin use, widely viewed as an important driver of the U.S. opioid crisis, the measure captures early geographic differences in misuse before subsequent changes in illicit-drug markets.¹⁰

Table 2 presents summary statistics for the baseline 2017 mortality measure, its 2010–2017 counterpart, and the three alternative exposure measures, together with their Spearman’s correlations with the baseline measure. All alternative measures are positively and significantly correlated with the age-adjusted death rate.

A potential concern with our exposure measure is that opioid exposure in 2017 may reflect pre-COVID labor-market conditions, suggesting that both cross-state differences in the post-COVID recovery of labor-force participation and opioid exposure could stem from shared pre-COVID factors. As indicated above, the interaction between opioids and labor-force participation can be complex, with effects going both ways ([Abraham and Kearney, 2020](#)). As shown in Figure 4a, there is no significant relationship between 2017 opioid exposure, measured by opioid-related deaths, and labor-force participation during the 2017–2019 period across states. Moreover, Figure 4b shows that the prime-age LFP of states with higher opioid exposure grew faster than that of lower exposure between 2015 and 2019.¹¹ There is

⁸Available at <https://www.deaiversion.usdoj.gov/arcos/arcos.html>. For details on opioid-treatment medications, see ([Mutter and Duchovny, 2022](#)). Naltrexone is excluded from ARCOS as it is not a scheduled drug.

⁹MME standardizes opioid potency relative to morphine, facilitating cross-opioid comparisons. Conversion is based on [Cutler and Glaeser \(2021\)](#).

¹⁰The measure is taken from the replication package for [Alpert, Powell, and Pacula \(2018\)](#), available at <https://www.openicpsr.org/openicpsr/project/114713/version/V1/view>. The measure calculates the share of the population above age 12 with non-medical OxyContin use. Population weights are used to aggregate shares across waves.

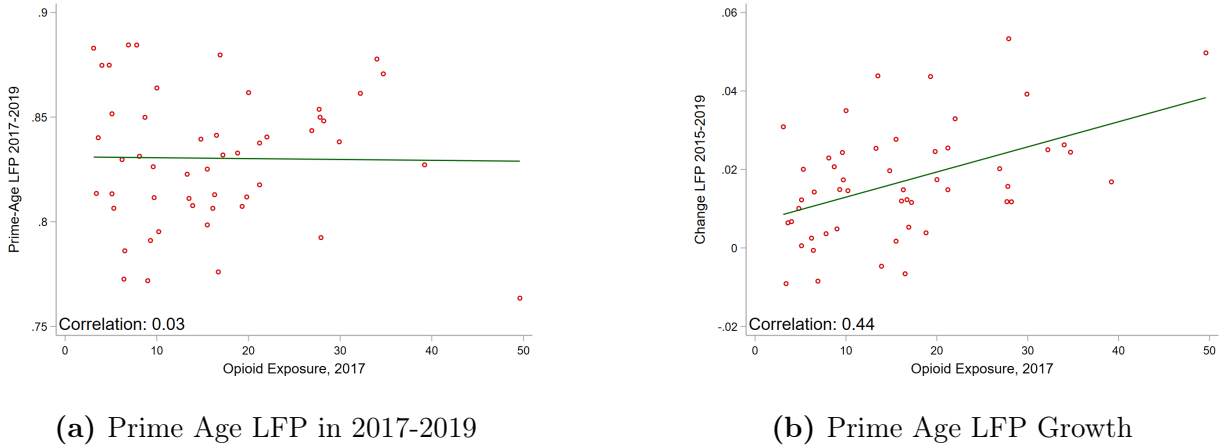
¹¹Figure A4 in Appendix A.3 analyzes this relationship using the three alternative measures of opioid

Table 2: Alternative Opioid Exposure Measures

	Mean	SD	Min.	Max.	Correlations
Death Rate, 2017	16.26	10.52	3.10	49.60	1.00
Death Rate, 2010-2017	10.83	5.75	2.90	33.90	0.91
Disorder, 2017-18 (%)	0.84	0.30	0.32	1.60	0.41
Treatment MME pc, 2017	925.43	601.09	157.06	2,809.07	0.78
Pre-2010 OxyContin misuse (%)	0.67	0.23	0.26	1.15	0.36

Notes: *Summary statistics for measures of state opioid exposure. The second row: age-adjusted opioid death rate for the combined period 2010-2017, obtained from CDC-MCOD. Third row: percentage of people with an opioid use disorder in 2017-2018, obtained from the NSDUH. Fourth row: quantity of drugs to treat opioid use disorder distributed to the state in 2017, measured as MME per capita, obtained from ARCOS. Last row: pre-2010 OxyContin exposure, taken from [Alpert, Powell, and Pacula \(2018\)](#). The last column shows Spearman's rank correlation between these measures and the 2017 age-adjusted death rate.*

no significant correlation between the measures and LFP between 2017 and 2019. Hence, the COVID-19 shock reversed an upward LFP trend in high-exposure states rather than reinforcing a downward one.

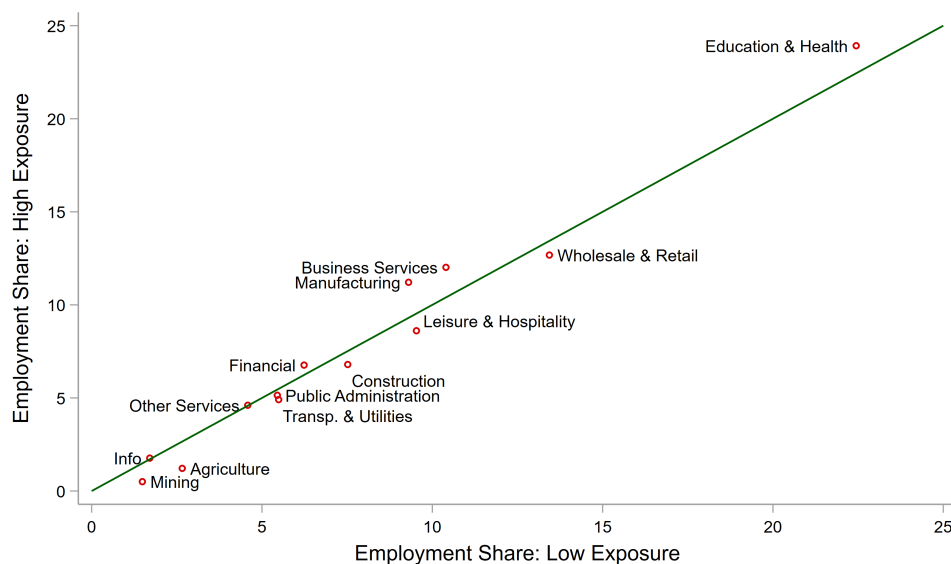
Figure 4: Relationship Between Opioid Exposure and Prime Age Labor Force Participation

Notes: *Left: Correlation between the average prime-age LFP between 2017 and 2019 and the age-adjusted opioid-related overdose death rate for 2017. Right: Correlation between prime-age LFP growth between 2015 and 2019 and the age-adjusted opioid-related overdose death rate for 2017. Source: CDC-MCOD and CPS.*

A further concern is that post-COVID labor-force participation differences across states exposure.

with higher or lower initial opioid exposure might reflect differences in the sectoral composition of employment. Although we control for industry composition in the main analysis, we show here that high- and low-exposure states had broadly similar employment structures.¹² As shown in Figure 5, employment shares across industries were very similar in high- and low-exposure states. Indeed, high-exposure states had a slightly lower share of employment in the two sectors most adversely affected by COVID-19: leisure and hospitality, and wholesale and retail trade. Moreover, Figure A5 in Appendix A.3 highlights that there were no diverging industry composition trends before 2020.

Figure 5: Industry Shares in High and Low Opioid Exposure States



Notes: *Employment shares across major industries in 2017 for states grouped according to whether opioid exposure is at or above the median or below it, where exposure is defined using the 2017 age-adjusted opioid death rate. Shares sum to 100 percent within each group. Source: CPS.*

Finally, the empirical analysis also includes control variables that reflect differences in COVID-19 intensity and policy responses across states. State-level COVID-19 cases and deaths are taken from the CDC’s COVID-19 Tracker.¹³ Policy data are obtained from the Oxford Tracker Dataset (Hale et al., 2021); specifically, the Stringency Index (covering school and workplace closures and stay-at-home orders) and the Economic Support Index (summarizing income support policies during COVID-19) are used.¹⁴ The average value of

¹²High exposure states are states with opioid exposure in 2017 above or equal to the median.

¹³Available at <https://covid.cdc.gov/covid-data-tracker/#datatracker-home>.

¹⁴The Oxford Tracker Dataset categorizes various COVID-19 policies into indexes representing different

these two indexes and their standard deviations across US states are shown in Figure A6 in Appendix A.4.¹⁵ Given the varied impact of COVID-19 across industries, the empirical analysis also includes a Bartik-style control, described below.

3 Empirical Strategy

The simultaneous impact of COVID-19 across all states makes a standard difference-in-differences estimator unsuitable for analyzing why some states experienced a slower recovery in labor-force participation. The analysis here uses an approach that allows the COVID-19 shock to have varying effects across states, depending on their pre-COVID opioid exposure. This approach relies on three assumptions: (1) the treatment affects all units (states) simultaneously, (2) the treatment effect depends on initial, predetermined conditions unrelated to the treatment, and (3) these initial conditions vary across units. The core idea is that states with higher pre-COVID opioid exposure experienced a slower recovery in labor-force participation, potentially due to greater availability of illicit drugs or a larger population with opioid-use experience, which became more salient post-COVID. Importantly, the COVID-19 onset is assumed to be independent of initial opioid exposure levels.

This empirical strategy is used by [Alpert, Powell, and Pacula \(2018\)](#) to study the effects of OxyContin’s reformulation, which made it harder to abuse. The reformulation, an event that affects everyone, is found to lead to greater increases in heroin deaths in states with higher pre-reformulation OxyContin exposure. A similar approach is also used by [Beheshti \(2023\)](#) to assess the impact of hydrocodone rescheduling by the Drug Enforcement Administration (DEA), which made it harder for doctors to prescribe. Areas with higher initial hydrocodone prescription rates show greater improvements in labor force participation and employment.

The empirical approach employs a dynamic two-way fixed effects model. The main dependent variable is the labor-force participation rate. The key explanatory variables are interactions between month indicators and pre-COVID opioid exposure. February 2020 is the omitted reference month, and March 2020 marks the beginning of the post-onset period. Pre-COVID exposure is measured in two ways using 2017 age-adjusted opioid death rates. The first measure is a binary variable categorizing states into "high" or "low" exposure groups

policy strengths, normalized between 0 and 100.

¹⁵[Ruhm \(2024\)](#) finds that if all states had imposed COVID-19 restrictions similar to those used in the 10 most restrictive states, excess deaths would have been 10 percent to 21 percent lower.

based on whether their death rates are at or above the median or below it. This method enables straightforward, difference-in-differences-like comparisons but sacrifices granularity. For example, Virginia, with 15 opioid deaths per 100,000, falls just below the median in the low-exposure group, while Utah, with 15.5 deaths per 100,000, is at the median in the high-exposure group alongside states like Ohio and West Virginia, which have much higher death rates (39 and 50 per 100,000, respectively). The second measure treats death rates as a continuous variable, preserving finer distinctions in exposure. To facilitate interpretation, this continuous variable is standardized to a mean of 0 and a standard deviation of 1.

The event study analysis is based on the following regression:

$$Y_{s,t} = \alpha_s + \alpha_t + \sum_{T \neq 2020m2} \beta_T \times \mathbb{1}\{t = T\} \times Op. Exp_s + \delta X_{s,t} + \varepsilon_{s,t}, \quad (1)$$

where $Y_{s,t}$ is the outcome of interest, LFP in state s at time t for the main analysis, expressed on a 0–100 scale so that coefficients are measured in percentage points. The coefficients α_s and α_t are the state and time fixed effects. The variable $Op. Exp_s$ is pre-COVID opioid exposure for state s , which can be either binary (1 if pre-COVID opioid exposure is at or above the median and 0 otherwise) or continuous. The variable $X_{s,t}$ represents time-varying controls.

The coefficients of interest in equation (1) are the β_T , with $\beta_{Feb.2020}$ normalized to 0. These coefficients represent changes in the outcome variable $Y_{s,t}$ relative to the month before COVID-19, across different levels of prior opioid exposure. For instance, when $Op. Exp_s$ is the binary exposure measure, a value of $\beta_{March.2020} = -1$ indicates that the change in labor-force participation from February to March 2020 was 1 percentage point lower in high-exposure states than in low-exposure states, conditional on the included fixed effects and controls. For the continuous exposure measure, $\beta_{March.2020}$ captures the difference in this change associated with a one-standard-deviation increase in the 2017 age-adjusted opioid death rate.

The control vector $X_{s,t}$ includes monthly COVID-19 case rates (set to zero before March 2020) to capture cross-state differences in pandemic severity. We also include the Oxford Stringency Index and the Economic Support Index, which summarize state-level containment and income-support policies that may directly affect labor supply and economic activity. These variables account for contemporaneous differences in public-health conditions and

policy responses across states.¹⁶ Lastly, following [Di Maggio and Kermani \(2016\)](#), we include a Bartik-style control to account for differences in states’ pre-pandemic industry composition. The control is constructed as

$$B_{s,t} = \sum_k w_{s,k,2017} \left(\frac{\nu_{-s,k,t} - \nu_{-s,k,t-1}}{\nu_{-s,k,t-1}} \right),$$

where $w_{s,k,2017}$ is industry k ’s share of employment in state s in 2017, and $\nu_{-s,k,t}$ is industry k ’s share of total national employment at time t , calculated after excluding state s . The state-industry employment shares are constructed from CPS respondents aged 18–64 who report being employed. Thus, the control assigns different predicted employment shocks to states according to their initial industry composition and subsequent national changes in industry employment shares.

The identifying assumption underlying equation (1) is that, absent the COVID-19 pandemic, states with different levels of pre-pandemic opioid exposure would have experienced parallel trends in prime-age labor-force participation. This parallel trends assumption allows post-COVID differences to be interpreted as difference-in-differences outcomes. Consequently, the estimated values of β_T should be close to 0 during the pre-COVID period. However, this assumption may be too strong and could be violated in the data. Following [Beheshti \(2023\)](#) and [Dobkin et al. \(2018\)](#), a more flexible specification, which allows for preexisting differential linear time trends across states, is also considered, given by

$$Y_{s,t} = \tilde{\alpha}_s + \tilde{\alpha}_t + \sum_{T > 2020m2} \theta_T \mathbb{1}\{t = T\} \times Op. Exp_s + \sum_S \phi_S \mathbb{1}\{s = S\} \times t + \tilde{\delta} X_{s,t} + \varepsilon_{s,t}, \quad (2)$$

where the term $\sum_S \phi_S \mathbb{1}\{s = S\} \times t$ is the term that allows for state-specific linear time trends. The identifying assumption no longer requires parallel trends across states; instead, it relies on similar deviations from potentially different linear trends in each state. In this specification, the coefficients θ_T are estimated only for the post-COVID period. However, the β_T estimates from equation (1) can be used to derive θ_T for the pre-COVID period ([Dobkin et al., 2018](#)). If these constructed pre-COVID estimates are close to 0, the linear trend assumption is a good fit to the data. Following [Beheshti \(2023\)](#), equation (1) is referred to as a non-parametric event study specification, while equation (2) is called a parametric event study specification. Both event-study regressions use population weights and standard

¹⁶Conditioning on post-onset variables may raise concerns if pandemic severity or policy responses lie on the causal path between pre-pandemic opioid exposure and labor-force participation, or if they induce collider bias. We therefore report the main results both with and without COVID-related controls.

errors clustered by state.

4 Results

4.1 Opioid Exposure and Deaths

The empirical strategy builds on the hypothesis that the COVID-19 shock generated differential increases in illicit drug use across states, depending on their pre-COVID opioid exposure, and that these differences may have contributed to subsequent divergence in labor-force participation. More specifically, we expect states with higher pre-COVID opioid exposure to experience larger increases in opioid-related mortality following the COVID-19 shock, or more broadly, higher drug overdose mortality.¹⁷ To examine this pandemic-era overdose response, we estimate equation (1) using the drug overdose death rate as the dependent variable, $Y_{s,t}$, and the continuous age-adjusted opioid death rates as our measure of pre-COVID opioid exposure, $Op. Exp_s$.¹⁸ In this overdose analysis, our objective is to isolate the effect of opioid exposure on illicit drug use, proxied by drug overdose deaths. For this reason, we include an additional control: the share of deaths involving fentanyl. Fentanyl is more lethal than other opioids. Moreover, its share of overdose deaths increased more rapidly between 2018 and 2023 in low-exposure states than in high-exposure states (see Figure B3 in Appendix B.1). Consequently, this control helps account for differential lethality that is not directly related to differences in drug use.¹⁹

The findings, shown in Figure 6, reveal that a one-standard-deviation increase in initial opioid exposure is associated with an increase of 0.46 deaths per 100,000 residents in the monthly crude overdose death rate by May 2020 and an increase of 1.74 deaths per 100,000 residents in the annual age-adjusted overdose death rate in 2020. These effects are substantial and persistent, representing 24 percent of the monthly and 8 percent of the yearly pre-pandemic death rates. Only the non-parametric event study results are presented, as the

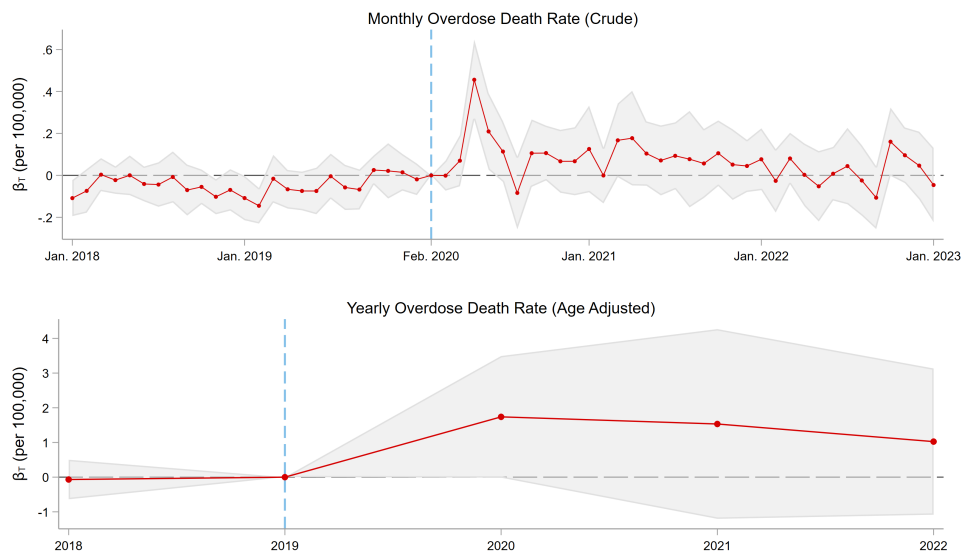
¹⁷For drug overdose deaths, the following ICD-10 codes are included: X40-X44 (accidental drug poisonings), X60-X64 (intentional self-poisoning by drugs), X85 (assault by drug poisoning), and Y10-Y14 (drug poisoning of undetermined intent).

¹⁸For results using binary exposure, where states are categorized as having 2017 age-adjusted death rates at or above versus below the median, see Figure B1 in Appendix B.1. For results using opioid overdose deaths, rather than all overdoses, see Figure B2 in Appendix B.1.

¹⁹For results without the fentanyl share control, see Figures B4 and B5 in Appendix B.1.

pre-COVID coefficients β_T do not indicate the presence of pre-trends. Furthermore, Figure B6 in Appendix B.1 shows the pandemic-era overdose-response results using the opioid death rate for the combined 2010-2017 period, and the three alternative opioid exposure measures described in Section 2. Across the four exposure measures, the estimates indicate a significant and persistent increase in overdose death rates in states with higher initial opioid exposure following the onset of COVID-19.

Figure 6: Non-Parametric Event Study with Continuous Exposure—Drug Overdose Deaths



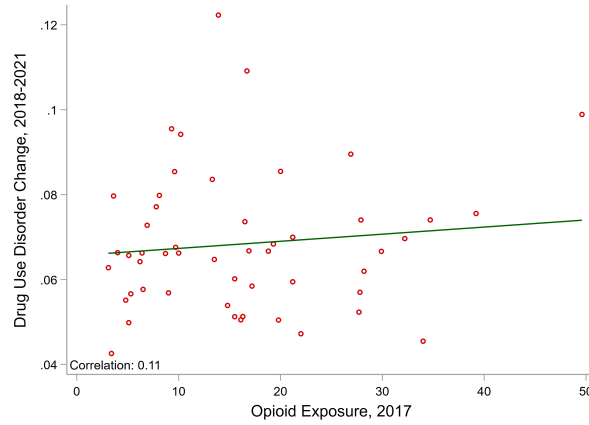
Notes: *Non-parametric event-study coefficients β_T and 95% confidence intervals. Coefficients represent changes in drug-overdose death rates between states with different levels of prior opioid exposure relative to the relevant reference period. The top panel reports the monthly crude drug-overdose death rate, for which February 2020 is the omitted reference month. The bottom panel reports the annual age-adjusted drug-overdose death rate, for which 2019 is the omitted reference year. In both panels, coefficients are measured in deaths per 100,000 residents. The initial opioid-exposure measure has been normalized to have a standard deviation equal to 1.*

An alternative way to highlight the hypothesis that the COVID-19 shock generated differential increases in illicit drug use across states would be to redo the analysis in Figure 6 using indicators for illicit drug use as the dependent variable. Such an analysis is not feasible since direct measures of illicit drug consumption are not available at the state-month level. The closest proxy comes from the National Survey on Drug Use and Health (NSDUH), which, as described in Section 2, reports the share of individuals with an illicit drug use disorder

for pooled multi-year periods (e.g., 2018-2019).²⁰

Furthermore, the 2020 NSDUH survey, conducted during the COVID-19 pandemic, was administered online and is not directly comparable with the 2019 survey. Still, we can compare the prevalence of drug use disorder before COVID-19 (2018–2019) and after the onset of the pandemic (2021–2022) and examine whether the increase is larger in states with higher pre-COVID opioid exposure. The results are shown in Figure 7. Consistent with our hypothesis, states with higher initial opioid exposure exhibit a larger increase in the share of individuals with a drug use disorder.

Figure 7: Relationship Between Opioid Exposure and Changes in Drug Use Disorder



Notes: *Correlation between the age-adjusted opioid-related overdose death rate for 2017 and the change in drug use disorder. The change is computed as the difference in the drug use disorder rate using the Restricted NSDUH data for the period 2018-2019 and 2021-2022. Source: CDC-MCOD and NSDUH.*

One identifying concern is that pandemic intensity or state policy responses may have been systematically related to pre-COVID opioid exposure. Figure B7 in Appendix B.2 examines this possibility by using monthly reported COVID-19 cases and deaths per 100,000 residents, the Stringency Index, and the Economic Support Index as outcomes in the non-parametric event-study specification in equation (1). A consistently positive or negative

²⁰Measures of illicit drug use disorder constructed from the NSDUH are not fully consistent across waves due to changes in diagnostic definitions. For 2018–2019, we use state-level estimates of past-year illicit drug use disorder based on DSM-IV criteria, which combine abuse and dependence. For 2021–2022, we use state-level estimates of past-year drug use disorder based on DSM-5 criteria. Both measures capture clinically meaningful substance use disorders and are broadly comparable indicators of severe illicit drug misuse. Data are obtained from the Interactive NSDUH State Estimates portal: <https://datatools.samhsa.gov/saes/state>

sequence of coefficients would indicate systematic differences in pandemic conditions between states with different levels of pre-COVID opioid exposure. The coefficients for cases and deaths are centered around zero. The policy coefficients are mostly insignificant and do not indicate that more-exposed states adopted systematically more stringent containment policies or more generous economic support. Thus, the figure provides little evidence that pre-COVID opioid exposure was systematically related to pandemic intensity or the policy responses considered.

4.2 Opioid Exposure and the Recovery of Labor-Force Participation

The results for labor-force participation are presented next.²¹ The β_T estimates from the non-parametric event study are shown in Figure 8. Because LFP is expressed on a 0–100 scale, the coefficients are directly measured in percentage points. The upper panel presents results from equation (1) using a binary measure of exposure, categorizing states according to whether their 2017 age-adjusted opioid death rate is at or above the median or below it.²² The lower panel shows results with the continuous measure of opioid exposure.

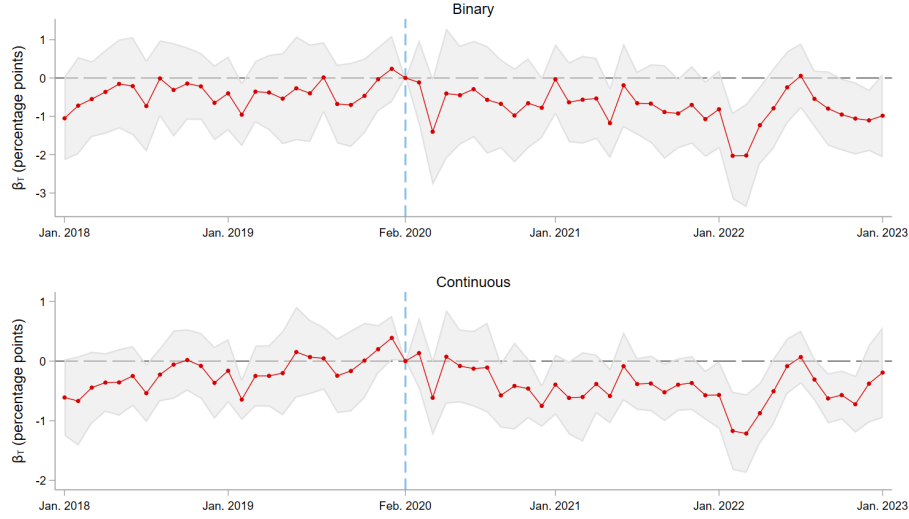
In both cases, the estimated effects are primarily negative, suggesting lower labor-force participation post-COVID in states with higher pre-COVID opioid exposure. However, these estimates are generally not statistically significant, except for February and March 2022. Additionally, the predominantly negative β_T coefficients during the pre-COVID period indicate differential pre-trends in prime-age labor force participation.

The estimates from the parametric event study, based on equation (2) and accounting for state-specific linear trends, are presented in Figure 9. The upper panel displays results using the binary measure of opioid exposure, while the lower panel shows results using the continuous measure. For both specifications, the pre-COVID estimated coefficients are approximately 0, supporting the identifying assumption that states experienced similar deviations from linear trends. For this reason, this is our preferred specification.

²¹Unlike the overdose analysis, the LFP regressions do not control for fentanyl share. Changes in fentanyl composition after the pandemic began may themselves be part of the pathway linking pre-COVID opioid exposure to labor supply, so conditioning on them could absorb part of the relationship of interest.

²²States classified as having pre-COVID opioid exposure at or above the median are: Connecticut, Delaware, District of Columbia, Florida, Illinois, Indiana, Kentucky, Maine, Maryland, Massachusetts, Michigan, Missouri, New Hampshire, New Jersey, New Mexico, New York, North Carolina, Ohio, Pennsylvania, Rhode Island, South Carolina, Tennessee, Utah, Vermont, West Virginia, and Wisconsin.

Figure 8: Non-Parametric Event Study, Labor-Force Participation



Notes: *Non-parametric event study coefficients β_T and the 95% confidence intervals. Coefficients represent changes in prime-age LFP relative to February 2020 between states with different levels of prior opioid exposure. For the top panel, the initial opioid exposure measure has been dichotomized: states with an age-adjusted opioid death rate in 2017 at or above the median are given a value of 1, and the others a value of 0. For the bottom panel, the initial opioid exposure measure has been normalized to have a standard deviation equal to 1.*

Post-COVID, the estimated θ_T coefficients indicate that labor-force participation rates in states with higher pre-COVID opioid exposure consistently fell below their linear trend compared to less-exposed states. The gap in labor-force participation between states with higher and lower opioid exposure emerged immediately after COVID-19. It widened, reaching nearly 1 percentage point by January 2021 and approximately 2 percentage points by January 2022. The upper panel estimates indicate that from March 2020 to January 2023, LFP in states with at-or-above-median opioid exposure averaged 1.10 percentage points further below its state-specific trend than in below-median states.

Similarly, the continuous measure in the lower panel shows that a one-standard-deviation increase in 2017 age-adjusted opioid death rates corresponds to an additional drop of 1.01 percentage points in LFP during the post-COVID period.²³ This implies that moving from the 25th percentile of initial opioid exposure (e.g., Minnesota) to the 75th percentile (e.g., Pennsylvania or Michigan) leads to an average LFP decline of 1.29 percentage points—a

²³The full-post-period average is estimated directly from the state-trend regression and accounts for the covariance among the 35 monthly coefficients (state-clustered standard error 0.26).

substantial effect, given that the standard deviation in prime-age LFP across states during the post-COVID period was 3.42 percentage points.²⁴

How do these magnitudes compare with earlier studies on the labor-supply effects of opioids? [Cho et al. \(2021\)](#) find that a one-standard-deviation increase in pre-2010 oxycodone exposure is associated with roughly a 0.3-percentage-point decline in both the prime-age employment-to-population ratio and the labor force participation rate. [Park and Powell \(2021\)](#) estimate that a 10 percent increase in exposure to the OxyContin reformulation—about one quarter of a standard deviation in their measure—reduces the employment-to-population ratio by approximately 0.14 percentage points. While our estimates are not directly comparable, as we use opioid-related mortality rather than prescription-based exposure, they are somewhat larger in magnitude but remain broadly consistent with these earlier findings.²⁵

4.3 The Effect of Overall Health

A potential concern with the analysis is that states with higher initial opioid exposure may also have weaker healthcare systems or poorer overall health. If this were the case, it could imply that the slower LFP recovery in more exposed states is not associated with higher exposure to the opioid epidemic but simply with worse overall health.

To address this concern, we conducted a placebo analysis by replacing initial pre-COVID opioid exposure in the main analysis with age-adjusted death rates from non-opioid-related causes. If these placebo results are significant, it would suggest that the larger drop in labor-force participation in states with higher opioid exposure may be attributed to broader health problems in the population or state-specific characteristics of the healthcare sector that influence both pre-COVID opioid mortality and post-COVID labor supply decisions.

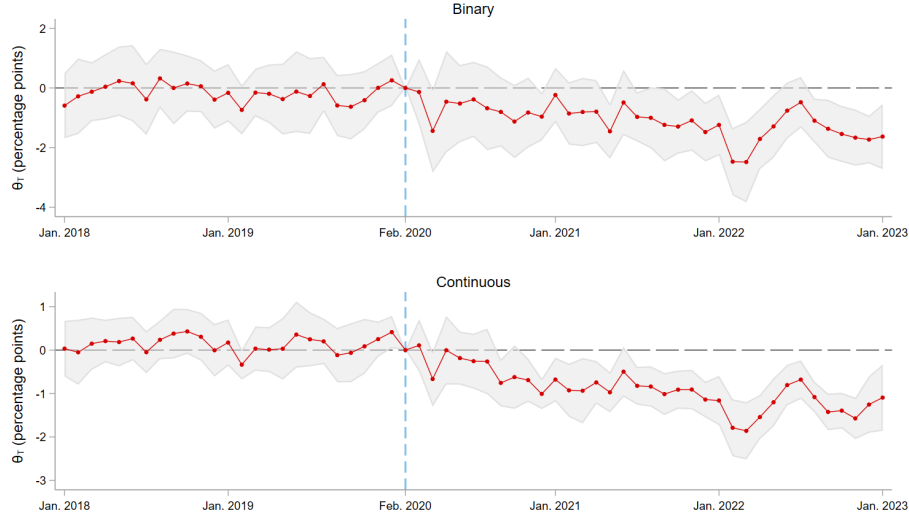
For the placebo analysis, we use data from the CDC Underlying Cause of Death dataset on the 15 leading causes of death in the United States.²⁶ Table B1 in Appendix B.3 lists

²⁴The dynamic pattern is similar when all time-varying controls are omitted, when only new monthly reported COVID-19 cases are omitted, or when cases are replaced with new monthly COVID-19 deaths. See Figure B8 in Appendix B.2.

²⁵Using pre-2010 OxyContin exposure in our robustness analysis yields an estimated decline in labor force participation of about 0.45 percentage points per one-standard-deviation increase, closely aligned with the magnitudes reported in prior work.

²⁶<https://wonder.cdc.gov/ucd-icd10.html>.

Figure 9: Parametric Event Study, Labor-Force Participation



Notes: *Parametric event study coefficients θ_T and the 95% confidence intervals. Coefficients represent average differences from state trends in prime-age LFP between states with different levels of prior opioid exposure. In the top panel, the initial opioid exposure measure has been dichotomized: states with an age-adjusted opioid death rate in 2017 at or above the median are given a value of 1, and the others a value of 0. For the bottom panel, the initial opioid exposure measure has been normalized to have a standard deviation equal to 1.*

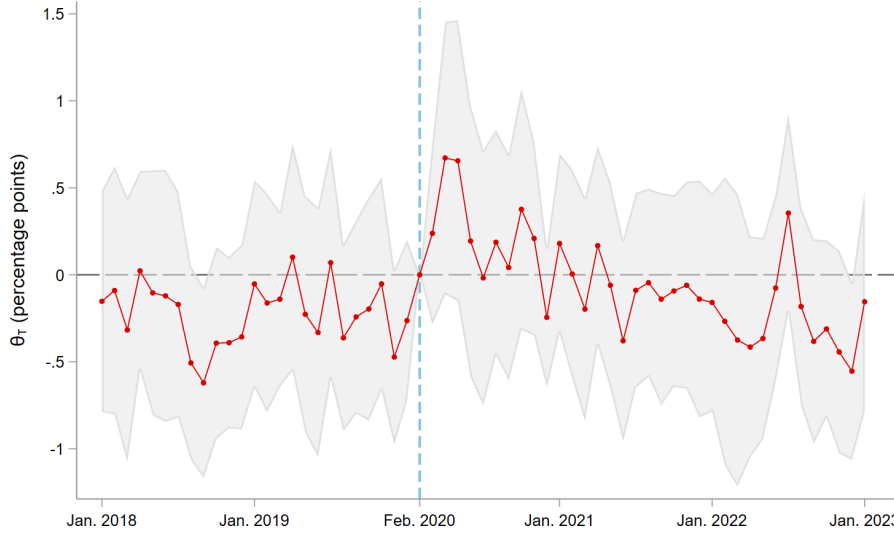
these causes along with their age-adjusted death rates for 2017. Deaths due to accidents and intentional self-harm (causes 3 and 8) may overlap with opioid-related deaths, and hence are excluded.²⁷ Figure 10 presents the estimated coefficients from the parametric event study where the initial exposure measure is based on the total death rate from the remaining 13 causes.²⁸ The coefficients for the post-COVID period are generally not statistically significant.²⁹

²⁷Accidents include underlying causes with ICD-10 codes X40-X44, and intentional self-harm includes ICD-10 codes X60-X64.

²⁸Using the non-parametric event study specification leads to similar results.

²⁹Results using the top 10 leading causes of death excluding accidents and intentional self-harm deaths, and each of the 13 underlying causes as the initial exposure measure, are shown in Figures B9 and B10 in Appendix B.3.

Figure 10: Parametric Event Study with Placebo Exposure



Notes: *Placebo parametric event study coefficients θ_T and the 95% confidence intervals when using prime-age LFP as the dependent variable. Coefficients represent average differences from state trends in prime-age LFP between states with different levels of prior placebo exposure. The measure of prior placebo exposure is the total age-adjusted death rate in 2017 due to the top 15 underlying causes of death, after excluding accidents and intentional self-harm. The death rate has been normalized to have a standard deviation equal to 1.*

4.4 Heterogeneity

The results for different demographic and socioeconomic groups are presented using the parametric event study with a continuous exposure measure, as specified in equation (2). Effects across gender and age groups are shown in Figure 11. The estimates are similar for men and women but show stronger effects among older individuals (ages 45–54) compared to younger ones (ages 25–44). Specifically, a one-standard-deviation increase in pre-COVID opioid exposure is associated with a post-period decline of 1.30 percentage points in labor-force participation for the 45–54 age group, compared to 0.85 percentage points for the 25–44 age group. Results for age groups 18–24 and 55–64, which are generally not statistically significant, are shown in Figure B11 in Appendix B.4.³⁰

³⁰In principle, the analysis in the previous section can be replicated at the county level. However, the primary data source, the Bureau of Labor Statistics’ Local Area Unemployment Statistics (LAUS), provides only an aggregate measure of the labor force. Hence, it is not possible to focus the analysis on prime-age workers. The results, while consistent with state-level analysis, are less precise, which is not surprising given our findings for different age groups.

Effects by educational attainment are illustrated in Figure 12 and show notable differences. For non-college-educated individuals, a one-standard-deviation increase in pre-COVID opioid exposure corresponds to a post-period decline of 1.24 percentage points in LFP relative to the linear trend. In contrast, the estimates for college graduates are not significantly different from zero. This result is in line with evidence that labor market outcomes for non-college-educated individuals are more affected by opioids (Aliprantis, Fee, and Schweitzer, 2023).

Figure 11: Parametric Event Study with Continuous Exposure

Labor-Force Participation by Gender and Age



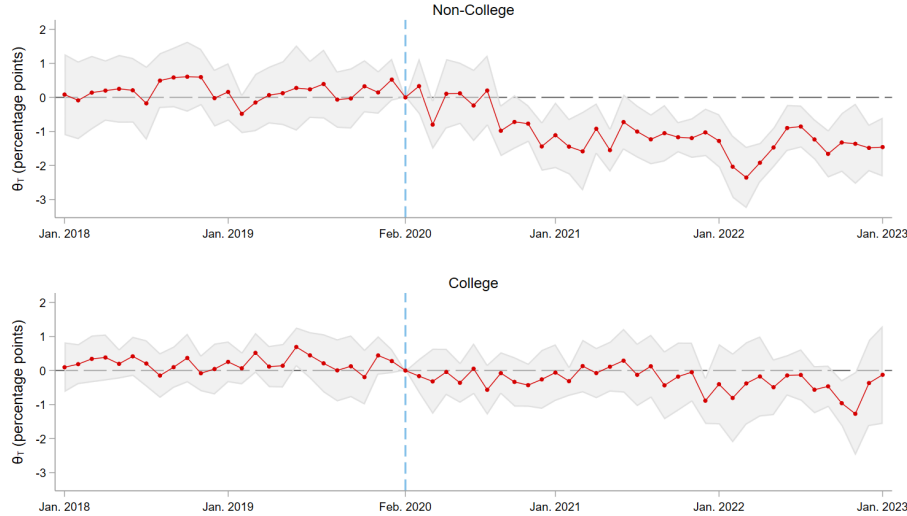
Notes: Parametric event study coefficients θ_T and the 95% confidence intervals. Coefficients represent average differences from state trends in prime-age LFP between states with different levels of prior opioid exposure. The top panel divides the sample by gender, displaying the prime-age labor force participation (LFP) among males and females. The bottom panels show LFP by age group: 25-44 on the left and 45-54 on the right. The initial opioid exposure measure has been normalized to have a standard deviation equal to 1.

5 Robustness

We conduct several sensitivity and robustness checks. First, we repeat the analysis after assigning placebo treatment dates before the pandemic. Second, we use alternative

Figure 12: Parametric Event Study with Continuous Exposure

Labor-Force Participation by Education



Notes: *Parametric event study coefficients θ_T and the 95% confidence intervals. Coefficients represent average differences from state trends in prime-age LFP between states with different levels of prior opioid exposure. The upper panel focuses on prime-age LFP for non-college-educated individuals, while the bottom panel focuses on college-educated individuals. The initial opioid exposure measure has been normalized to have a standard deviation equal to 1.*

measures of pre-COVID opioid exposure to assess the robustness of the results from our preferred parametric event-study specification in equation (2). Finally, we employ alternative empirical approaches to examine whether modeling choices drive the results.

5.1 Placebo Treatment Dates

We conduct a timing falsification exercise by re-estimating the parametric event-study specification for each of 26 placebo treatment dates from January 2017 through February 2019. For each placebo date, the estimation window comprises the preceding 24 months and the 12 months beginning with the assigned onset. The 12-month horizon ensures that every placebo post-period ends before March 2020. None of the placebo dates yields a negative and statistically significant estimate. By contrast, applying the same specification to the actual pandemic onset in March 2020 yields an average 0.57-percentage-point decline in labor-force participation relative to state-specific linear trends over the following 12 months for a one-

standard-deviation increase in pre-COVID opioid exposure.³¹ Figure B12 in Appendix B.5 reports the 26 placebo estimates together with the March 2020 estimate.

5.2 Alternative Exposure Measures

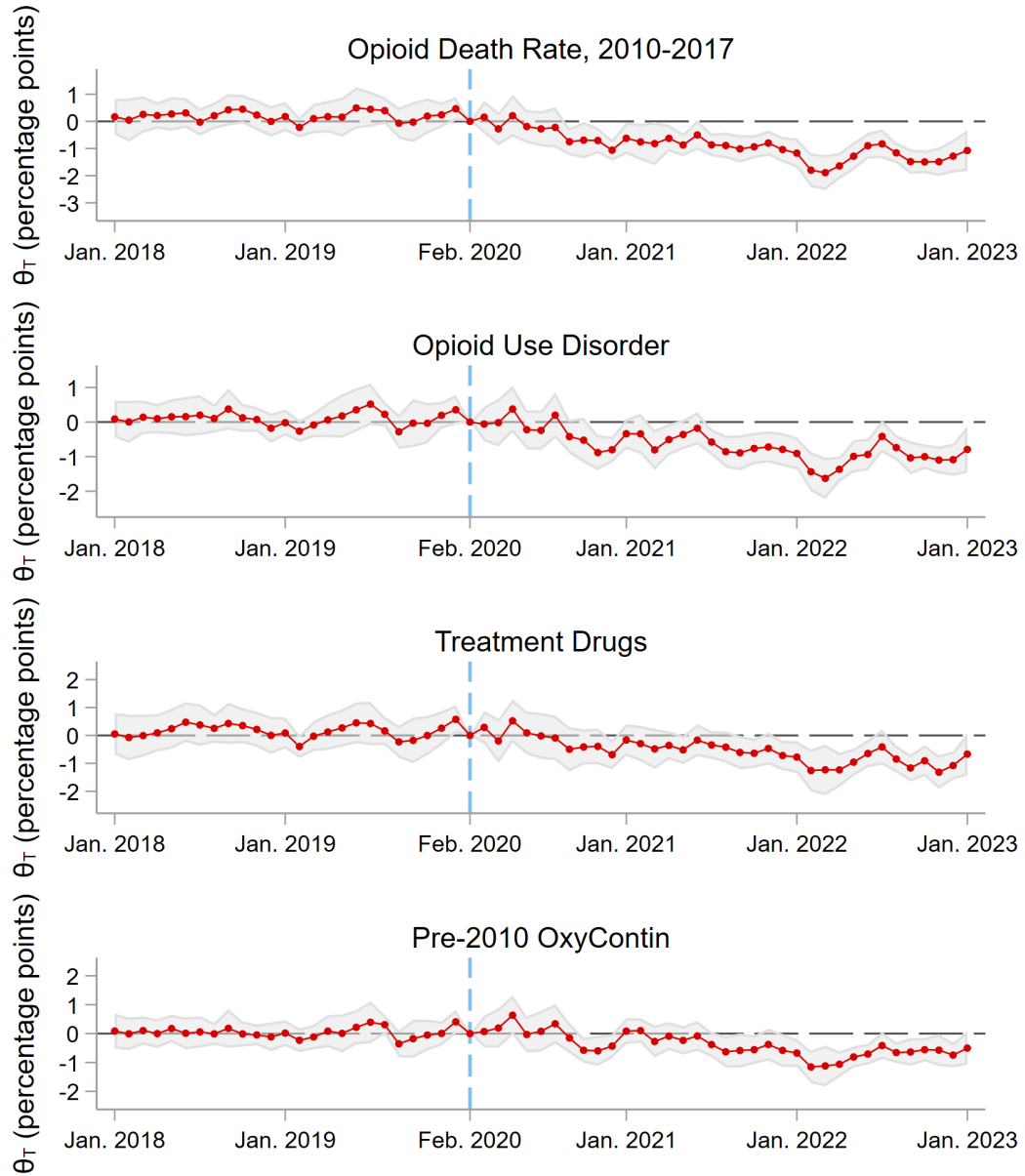
The analysis uses age-adjusted opioid death rates from 2017 as a measure of opioid exposure. In this section, we replace the single-year mortality measure with its 2010-2017 counterpart and consider three alternative measures of opioid exposure. The first alternative measure is the percentage of individuals aged 12 and above with an opioid use disorder, as reported in the NSDUH. The second is the per-capita shipment of medications for opioid addiction treatment, sourced from ARCOS. The third is pre-2010 OxyContin exposure, obtained from [Alpert, Powell, and Pacula \(2018\)](#).

Figure 13 reports estimates from the parametric event study using these continuous exposure measures. Although the magnitudes are somewhat smaller for the last three measures, the patterns are very consistent with those in Figure 9. A one-standard-deviation increase in the share of individuals with an opioid use disorder is associated with roughly a 0.76-percentage-point larger decline in labor-force participation relative to trend. For the per-capita supply of opioid treatment medications, the corresponding effect is about 0.67 percentage points. For pre-2010 OxyContin exposure, it is approximately 0.45 percentage points.³²

³¹The state-clustered standard error is 0.23. For this exercise, we extend the monthly CPS sample back to January 2015 and hold the baseline 2017 age-adjusted opioid death rate fixed across placebo dates. The placebo and March 2020 regressions omit COVID-19 and pandemic-policy controls, while retaining the employment-based Bartik control, so that the specifications are comparable across treatment dates. Because these estimates use a 12-month post-period, they are not directly comparable in magnitude with the main 35-month estimate of -1.01 percentage points. A 35-month post-period would extend most placebo windows into the COVID-19 period and leave substantially fewer eligible pre-pandemic treatment dates.

³²Given the slightly right-skewed distribution of 2017 death rates across states, estimates are also provided using the logarithm of death rates as the exposure measure. Figure B13 in Appendix B.6 presents results using these variants of pre-COVID opioid exposure in the parametric event-study specification with continuous exposure. The post-COVID gap in labor-force participation closely aligns with the estimates in Figure 9.

Figure 13: Parametric Event Study - Alternative Measures of Opioid Exposure



Notes: *Parametric event study coefficients θ_T and the 95% confidence intervals. Coefficients represent average differences from a state trend in prime-age LFP between states with different levels of prior opioid exposure. The top panel uses opioid death for 2010-2017. The second panel uses the percentage of people with opioid use disorder in 2017-2018. The third panel uses the amount of MME per capita of drugs to treat opioid use disorder distributed to the state in 2017. The bottom panel uses pre-2010 OxyContin exposure. In all four cases, the measure has been normalized to have a standard deviation equal to 1.*

5.3 Alternative Empirical Approaches

The parametric event study approach assumes that states with different initial levels of opioid exposure experience similar deviations from their state-specific linear trends. We assess this assumption in four ways. First, the parametric event study is re-estimated to allow for quadratic rather than linear state-specific trends. Second, the main analysis is repeated using a longer pre-period. Instead of estimating trends over 2018–2023 and using the 2017 opioid death rate as the exposure measure, the specification is re-estimated using data beginning in 2015 and using the 2015 opioid death rate as the measure of pre-COVID exposure. Figures B14 and B15 in Appendix B.7 show that, under these alternative specifications, the results remain consistent with the main findings: states with higher opioid exposure experience a persistently larger decline in LFP after COVID-19.

Next, we estimate the effects using a *synthetic control* approach, following Abadie and Gardeazabal (2003) and Abadie, Diamond, and Hainmueller (2010). For this method, we treat as treated those states with pre-COVID opioid exposure above or equal to the median value, and as controls the remaining states. Weights are assigned to the untreated states to construct a synthetic control group that closely matches the treated group’s pre-COVID labor force participation. These weights are time-invariant and are chosen to minimize the difference in pre-COVID LFP between the synthetic control and treated groups. By construction, the synthetic control reproduces the pre-COVID trend in LFP of the treated states. The treatment effect is then estimated as the post-COVID difference in average LFP between the treated group and its synthetic control.

Figure 14 illustrates the average prime-age labor-force participation in the treated states and in the synthetic control group. In the post-COVID period, LFP diverges between the two groups, with treated states—those with higher opioid exposure—showing systematically lower LFP. The average treatment effect over the post-COVID period is -0.59 percentage points.³³

Finally, we implement an alternative parametric event study proposed by Alpert, Powell, and Pacula (2018). This specification allows for a linear preexisting trend in LFP, a shift in its level at the onset of COVID-19, and a post-COVID trend break. Following the notation

³³The standard error is 0.31, based on 5,000 bootstrap replications resampling states with replacement and re-estimating the synthetic-control weights in each replication. The p -value is 0.054.

Figure 14: Synthetic Control and Treatment Group, Labor-Force Participation



Notes: Labor-force participation in treatment and synthetic control groups. The treatment group includes states with initial opioid exposure at or above the median. The synthetic control is a weighted average of states with below-median exposure, with weights chosen to reproduce the treated group's pre-COVID LFP path.

specified above, we have:

$$LFP_{s,t} = \alpha_s + \alpha_t + \delta_1[t \times Op. Exp_s] + \delta_2[Post_t \times Op. Exp_s] + \delta_3[Post_t \times (t - 2020m2) \times Op. Exp_s] + \delta X_{s,t} + \varepsilon_{s,t}, \quad (3)$$

where $Post_t = 1$ for $t \geq$ March 2020. The coefficients δ_1 , δ_2 , and δ_3 represent the pre-trend, the shift, and the post-trend, respectively. The variable $X_{s,t}$ includes the same state- and time-level controls as in the main analysis. Table 3 reports the results using our main opioid exposure measure in column 1 (the 2017 age-adjusted opioid death rate) and the three alternative exposure measures in columns 2–4. When using the main opioid exposure measure, the pre-trend coefficient is slightly positive, consistent with the non-parametric event study. Both the shift and the post-COVID trend-break coefficients are negative, indicating that high-exposure states experienced a persistent deterioration in LFP relative to low-exposure states. The cumulative effects after 6, 12, and 24 months closely match the magnitudes obtained in the baseline specification. Furthermore, for all three alternative opioid exposure measures, we obtain a significant post-trend coefficient of similar magnitude.

Table 3: Alternative Parametric Event Study, Labor Force Participation

	(1) Death Rate	(2) Disorder	(3) Treatment	(4) OxyContin
Pre-trend	0.0251*** (0.009)	0.0211*** (0.006)	0.0151 (0.164)	0.00986 (0.182)
Shift	-0.445* (0.094)	-0.210 (0.329)	-0.0948 (0.735)	0.0693 (0.767)
Post-trend	-0.0334*** (0.004)	-0.0319*** (0.001)	-0.0338** (0.027)	-0.0297*** (0.007)
Effect 6 Months	-0.645** (0.011)	-0.402* (0.063)	-0.298 (0.302)	-0.109 (0.637)
Effect 12 Months	-0.845*** (0.001)	-0.593** (0.011)	-0.501 (0.122)	-0.287 (0.243)
Effect 24 Months	-1.246*** (0.000)	-0.976*** (0.001)	-0.906** (0.042)	-0.644** (0.042)
Observations	3111	3111	3111	3111
Adj. R2	0.770	0.770	0.770	0.771

Notes: The table shows the estimates obtained from the regression equation (3) using different initial opioid exposure measures. Column 1 uses the age-adjusted opioid death rate. Column 2 uses the percentage of people with opioid use disorder in 2017-2018. Column 3 is the amount of MME per capita of drugs to treat opioid use disorder distributed to the state in 2017. Column 4: pre-2010 OxyContin exposure. The initial opioid exposure measure has been normalized to have a standard deviation equal to 1. *p*-values in parentheses, * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. State clustered S.E. used. States have been weighted by their population size.

6 Mechanisms

The COVID-19 pandemic coincided with an unprecedented rise in deaths from opioids and other illicit drugs, driven by multiple factors. Pandemic-induced isolation worsened mental health (Panchal et al., 2020; Adams-Prassl et al., 2022a), while disruptions in medical services reduced access to medications for opioid use disorder (Currie et al., 2021a; Russell et al., 2021). This period also saw a shift in opioid consumption toward fentanyl, a more potent opioid (Currie et al., 2021b). Additionally, increased free time, lower drug prices, and government income support may have exacerbated the use of illicit drugs (Mulligan, 2022).

Although drug overdose deaths rose nationwide during COVID-19, the impact was most severe in states with higher pre-COVID opioid exposure, as shown in Figure 6. This differen-

tial effect may reflect established illegal opioid distribution networks, more lenient prescribing practices, or a larger population experienced with opioids in highly affected states.

The larger increase in drug-overdose mortality in states with higher initial opioid exposure may have contributed to their slower recovery in labor-force participation. To provide descriptive evidence of this potential mechanism, Table 4 compares labor-market and health outcomes in 2019 for nonusers and individuals with a drug use disorder, aged 24–49. Individuals with a drug use disorder were more likely to be out of the labor force: 26 percent, compared with 17 percent among nonusers.³⁴

Individuals with a drug use disorder are more likely to report disability as the reason they are not participating in the labor market. This mechanism is consistent with evidence from Park and Powell (2021), who show that increased exposure to the OxyContin reformulation led to higher rates of disability-related nonemployment and worsening health among affected populations. As Table 4 reports, in 2019, 17 percent of nonparticipants who did not use any drugs reported disability as the reason compared with 25 percent of those with a drug use disorder. Drug use disorder is also associated with more missed workdays: those with drug use disorders missed an average of 4.1 workdays per month, compared to 1.1 days among nonusers. Self-reported health outcomes were similarly much poorer: 26 percent of individuals with drug use disorders reported fair or poor health, compared to 10 percent among nonusers. Finally, comparing 2019 data with the 2021–2022 period, Figure 15 shows that the LFP and disability gaps between nonusers and those with drug use disorders widened after COVID-19.³⁵

If greater drug use is one channel linking pre-COVID opioid exposure to the slower recovery in labor-force participation, we would expect states with higher pre-COVID opioid exposure to experience a larger increase in disability-related nonparticipation. This implication is consistent with the descriptive NSDUH evidence that drug use disorder is associated with disability-related nonparticipation. To examine this relationship across states, we use CPS information on respondents’ reasons for nonparticipation. Individuals who are not in the labor force report a main reason, including being “retired,” being “unable to work” because of a medical condition expected to prevent work for six months or more, or “other.” Respondents selecting “other” may further report disability, illness, school attendance, or

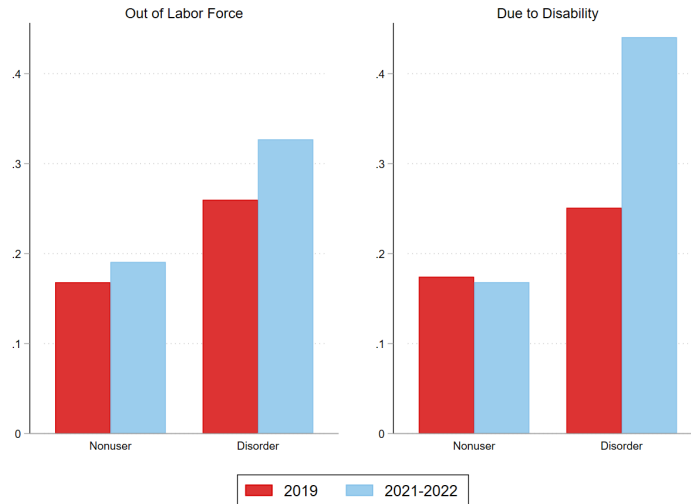
³⁴This age group is used because NSDUH reports age in brackets.

³⁵Tables C1 and C2 in Appendix C.1 provide detailed employment and health outcomes by drug use in 2019 and the pooled 2021–2022 period.

Table 4: Employment and Health by Drug Use, Ages 24-49

Panel A: Employment	Out of LF	Disability	Workdays Missed
Nonuser	17%	17%	1.1
Disorder	26%	25%	4.1
Panel B: Self-Reported Health	Very Good	Good	Fair/Poor
Nonuser	61%	29%	10%
Disorder	40%	35%	26%

Notes: Employment-related variables in Panel A, and self-reported health status in Panel B, by type of drug user in 2019. In Panel A, the first column shows the share of individuals not in the labor force. The second column shows the share of individuals out of the labor force who report disability as the reason for not participating. The third column displays the average number of workdays missed in the past 30 days due to illness or other reasons. Self-reported health is grouped into three categories: “Very Good” (including excellent and very good health), “Good,” and “Fair/Poor” (including fair and poor health). Nonusers did not use any drug (excluding marijuana) in the past 12 months. Disorder indicates a drug use disorder (excluding marijuana) in the last 12 months. Source: NSDUH.

Figure 15: Changes in Labor-Force Participation by Drug Use, Ages 24-49

Notes: Out of labor force variables by type of drug use in 2019 and 2021–2022. Left panel: share of individuals who are out of the labor force. Right panel: among individuals out of the labor force, the share who report disability as the reason for not participating. Nonusers did not use any drug (excluding marijuana) in the past 12 months. Disorder indicates a drug use disorder (excluding marijuana) in the last 12 months. Source: NSDUH.

care of house or family as the reason for nonparticipation.

Using this information, the prime-age out-of-labor-force due to disability rate is defined

as the share of prime-age individuals who do not participate in the labor force and have as their reason "unable to work" or "disability." The out-of-labor-force due to disability rate has been on a declining trend since 2015, driven mainly by those without a college degree. The decline, however, stopped with the pandemic, stabilizing at around 0.05 (Figure C1 in Appendix C.2).

To analyze the post-COVID trajectory across states of the share of prime-age individuals who are out of the labor force due to disability, a parametric event study using a continuous measure of initial opioid exposure, as specified in equation (2), is estimated with the out-of-labor-force due to disability rate as the dependent variable. The estimated coefficients are presented in Figure 16. The results indicate that pre-COVID, states with varying opioid exposure levels exhibited similar deviations from their state-specific trends. Post-COVID, however, states with higher preexisting opioid exposure experienced larger positive deviations from these trends. Specifically, a one-standard-deviation increase in opioid exposure is associated with a 0.43-percentage-point increase in the post-COVID out-of-labor-force due to disability rate. This is a substantial effect, given that the standard deviation of the out-of-labor-force due to disability rate during the post-COVID period was 1.96 percentage points.³⁶

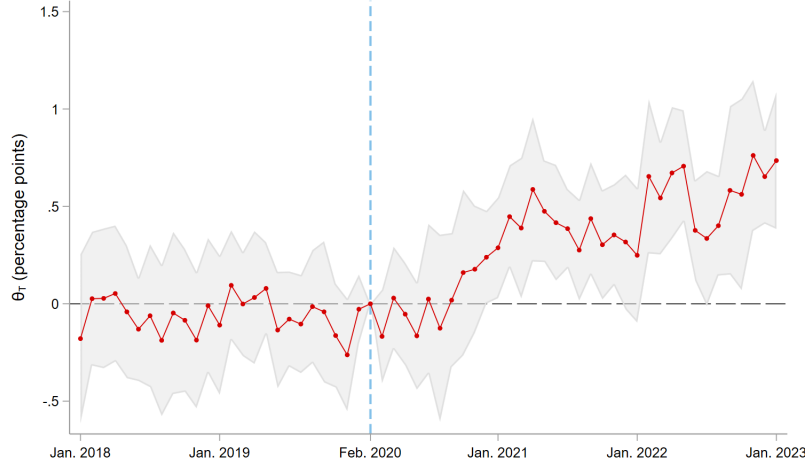
7 Conclusions

The COVID-19 pandemic caused substantial and persistent declines in labor force participation across the United States, particularly among prime-age individuals without a college degree. This paper shows that the decline was larger in states with greater pre-pandemic exposure to the opioid epidemic. States with higher opioid-related mortality before COVID-19 experienced markedly slower recoveries in labor-force participation following the pandemic shock. Our event-study estimates indicate that a one-standard-deviation increase in pre-pandemic opioid mortality is associated with an additional decline of 1.01 percentage points in the prime-age labor-force participation rate relative to trend.

Importantly, this relationship is not driven by broader differences in state health or economic conditions. We find no comparable association between non-opioid mortality and

³⁶As with labor-force participation, these results are primarily driven by the non-college-educated population, as shown in Figure C2 in Appendix C.3.

Figure 16: Parametric Event Study Estimates—Out of LF due to Disability



Notes: *Parametric event study coefficients θ_T and the 95% confidence intervals. Coefficients represent average differences from state trends in prime-age out-of-labor-force due to disability rates between states with different levels of prior opioid exposure. The out-of-labor-force-due-to-disability rate is defined as the proportion of individuals in the total population who are out of the labor force due to disability. The initial opioid exposure measure has been normalized to have a standard deviation equal to 1.*

post-pandemic labor outcomes, suggesting that generally poorer health alone cannot explain the weaker recoveries observed in high-opioid states. We further provide evidence on a potential mechanism: in these states, exits from the labor force due to disability or illness rose disproportionately after COVID-19. This pattern is consistent with individual-level evidence showing that individuals with drug use disorders are substantially more likely to be out of the labor force and to report health-related work limitations.

Taken together, our findings highlight how overlapping public health crises can interact to produce lasting labor market effects. More broadly, they suggest that the labor force consequences of future economic shocks may depend critically on preexisting public health vulnerabilities.

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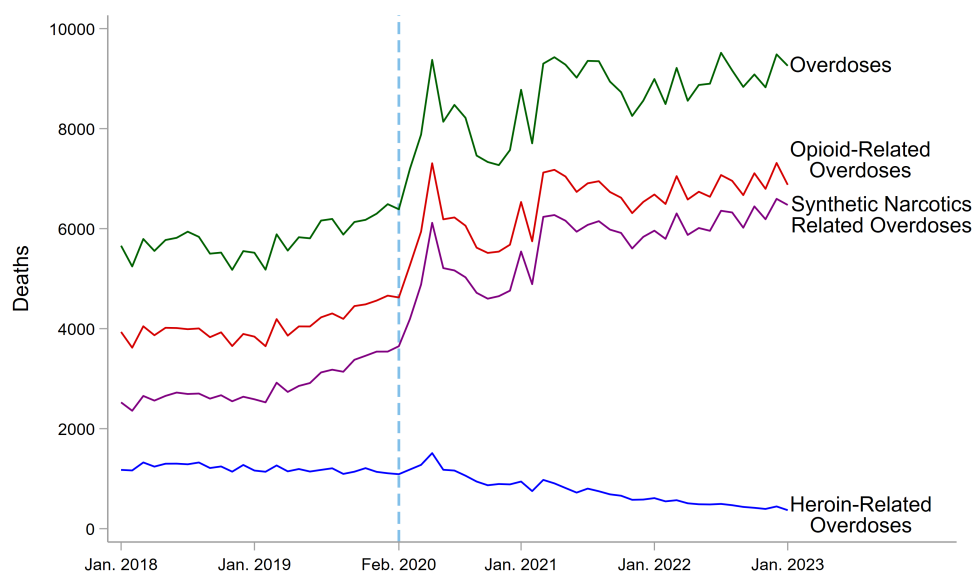
Online Appendix

A Data

A.1 Opioid Deaths Across the United States

Figure A1 shows monthly deaths from all overdoses, opioid-related overdoses, and specific opioid-related categories (synthetic opioids and heroin) from 2018 to 2023 in the United States. The figure highlights that the post-COVID rise in opioid-related mortality has been driven primarily by synthetic opioids.

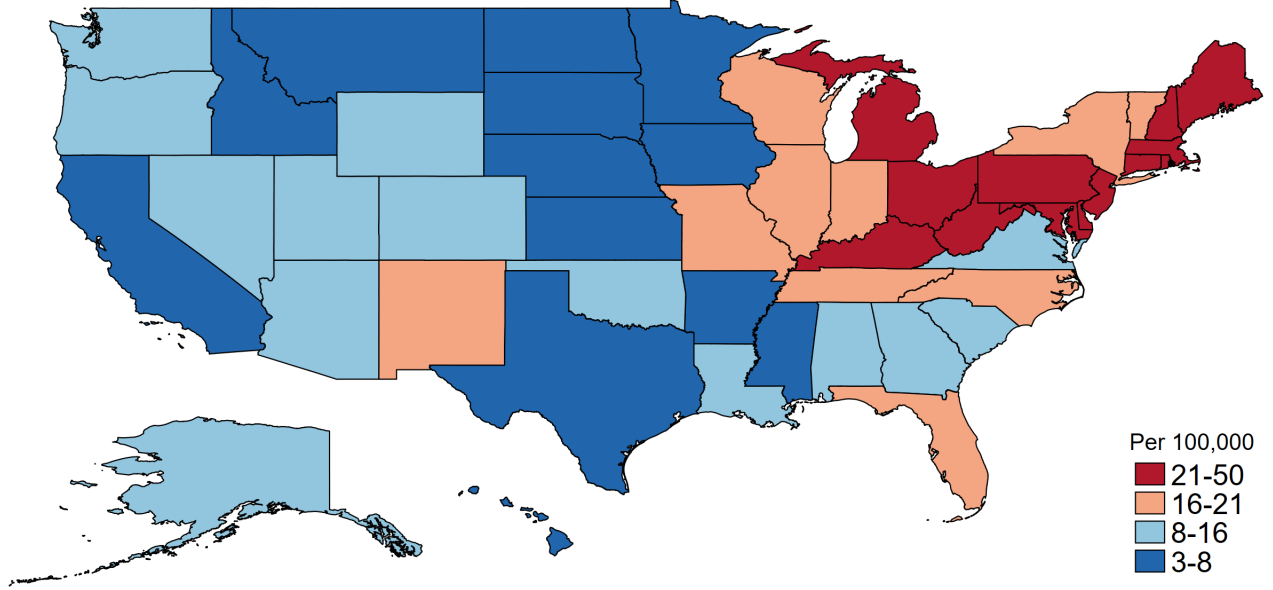
Figure A1: Monthly Overdose Deaths by Drug Type



Notes: Monthly deaths involving all overdoses, opioid-related overdoses, synthetic opioids, and heroin from 2018 to 2023. Source: CDC-MCOD.

Figure A2 shows the age-adjusted opioid-related overdose death rates in 2017 across the United States. The regions with the highest death rates, ranging from 16 to 50 per 100,000, are the Appalachian region, the Rust Belt, and New England.

Figure A2: Age-Adjusted Death Rates from Opioid Overdose, 2017



Notes: Age-adjusted opioid-related overdose death rate for 2017 across the US. Death rates are computed per 100,000 people. Source: CDC-MCOD.

A.2 Labor Force Participation Trends

To obtain the difference between the actual and predicted prime-age labor-force participation shown in Figure 3, we do the following steps. First, we estimate a linear time trend in prime-age LFP for each state from January 2015 to February 2020. Second, we use the estimated coefficients to forecast the prime-age LFP from March 2020 to December 2022. Lastly, we determine the difference between the observed values and the trend.

To predict the trend, we run the following state-level regression for the period January 2015 - February 2020:

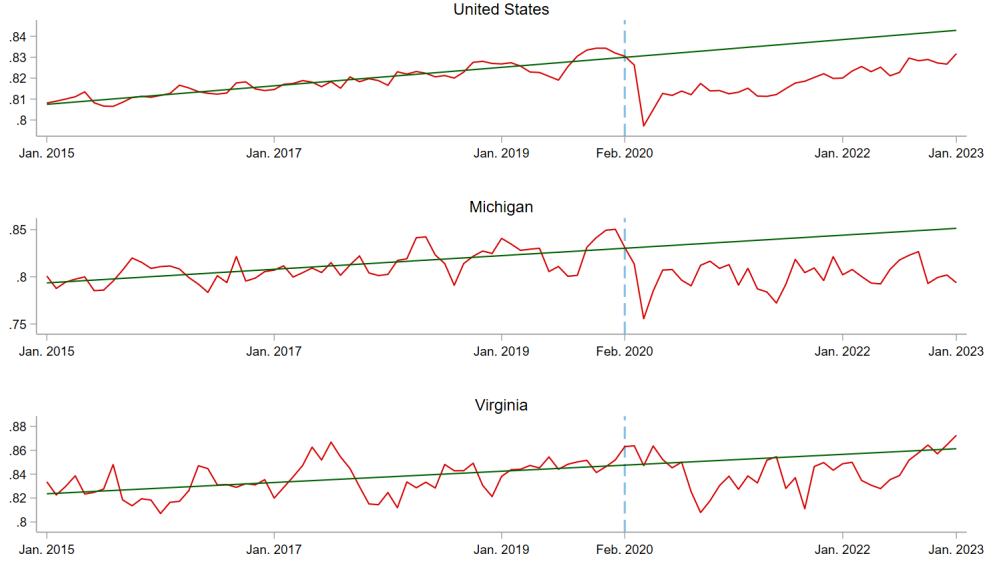
$$LFP_{i,t} = \alpha_i + \beta_i t$$

where $LFP_{i,t}$ is prime-age labor-force participation for state i at date t (monthly).

Using the estimated coefficients, we predict the prime-age LFP for each state for March 2020 to December 2022, $\widehat{LFP}_{i,t}$. Figure A3 shows the actual (red line), as well as the estimated and predicted (green line) prime-age LFP for the US, Michigan (a state where LFP stays below trend during the sample), and Virginia (where LFP recovers rather quickly), as

examples.

Figure A3: Estimated and Predicted LFP for the US, Michigan, and Virginia



Notes: Actual (red line) and predicted (green line) prime-age LFP. The figure shows the results for the entire United States and the states of Michigan and Virginia. Source: CPS.

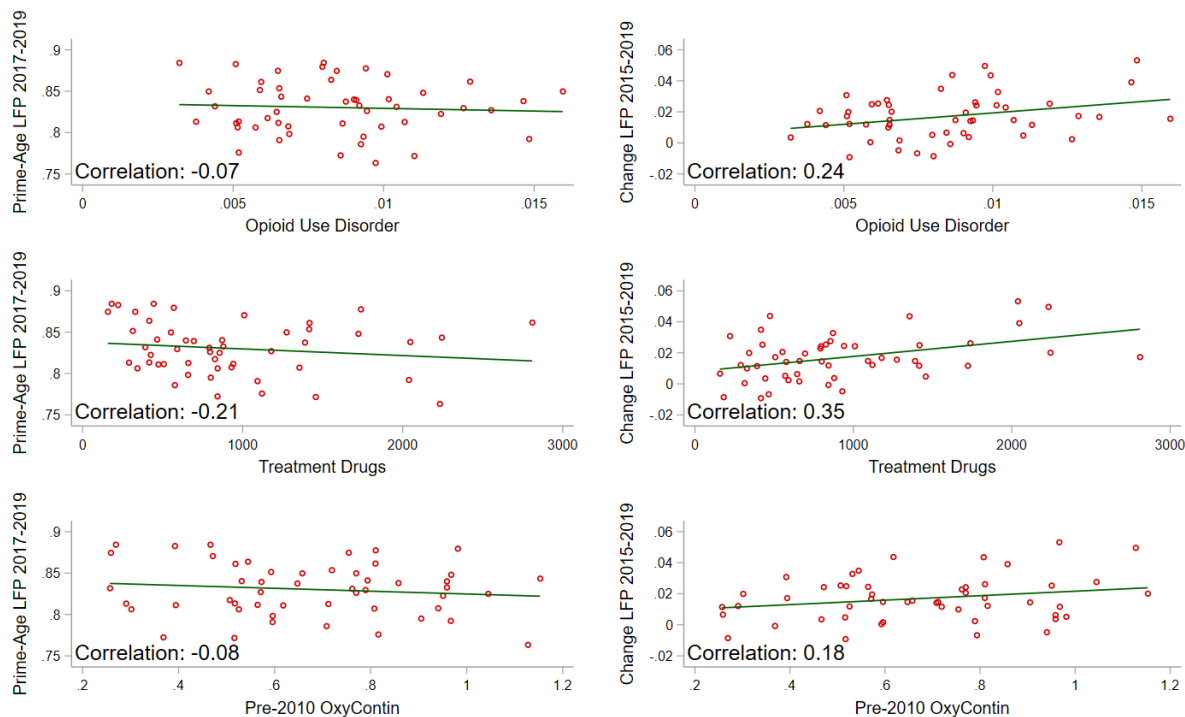
Then we obtain the difference as $Diff_{i,t} = LFP_{i,t} - \widehat{LFP}_{i,t}$. Negative values imply a lower labor-force participation than what has been predicted. To obtain the measure presented on the vertical axis of Figure 3, we average the difference for the period of July to December 2022.

A.3 Labor-Force Participation and Opioids

The left panels in Figure A4 show the correlations between alternative measures of opioid exposure and average prime-age LFP between 2017 and 2019. The right panels show the correlation between alternative measures of opioid exposure and the growth in prime-age LFP between 2015 and 2019. The alternative measures are: the percentage of individuals with opioid use disorder in 2017–2018 (first row), MME per capita shipments of medications used to treat opioid addiction (second row), and pre-2010 OxyContin exposure (third row). Across the three measures, the correlations with average prime-age LFP between 2017 and 2019 are not statistically significant (left panels). Meanwhile, there is a positive and statistically significant correlation between the first two alternative measures and growth in prime-age

LFP between 2015 and 2019.

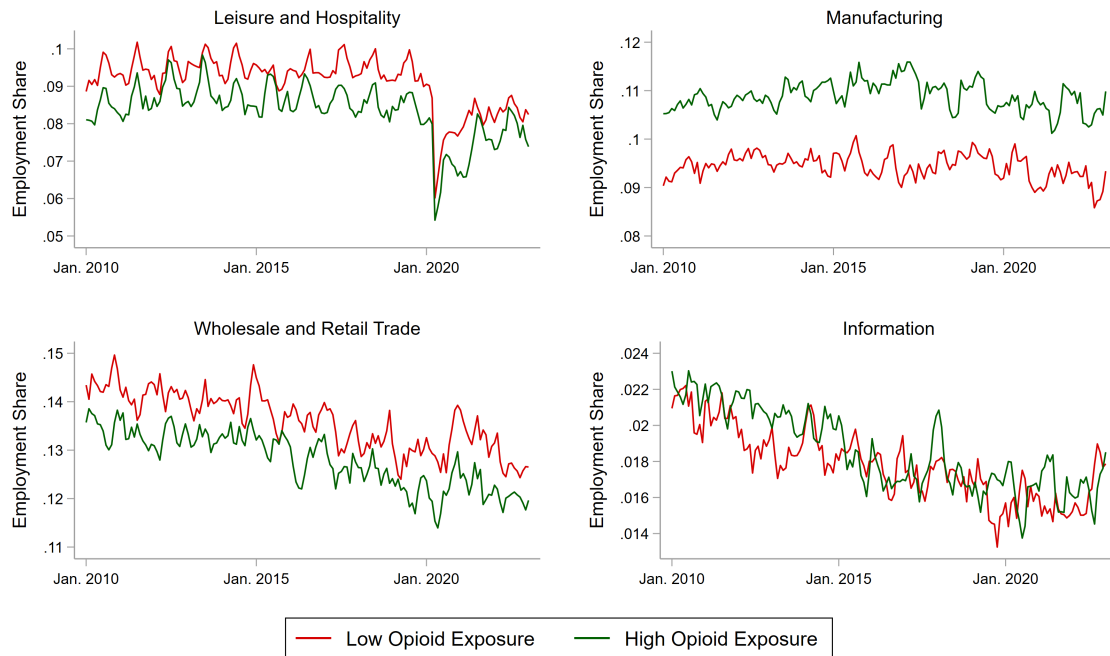
Figure A4: Correlations Between Prime-Age LFP and Alternative Opioid Exposure Measures



Notes: *Left panels: Correlations of alternative measures of opioid exposure with average prime-age LFP between 2017 and 2019. Right panels: Correlations of alternative measures of opioid exposure with prime-age LFP growth between 2015 and 2019. Source: CPS, NS-DUH, ARCOS.*

Figure A5 complements this by showing the time series of industry shares for selected sectors. The trajectories for both groups are parallel in the pre-COVID period, indicating no diverging industry composition trends prior to 2020.

Figure A5: Industry Employment Shares Over Time: Selected Sectors, 2010–2023

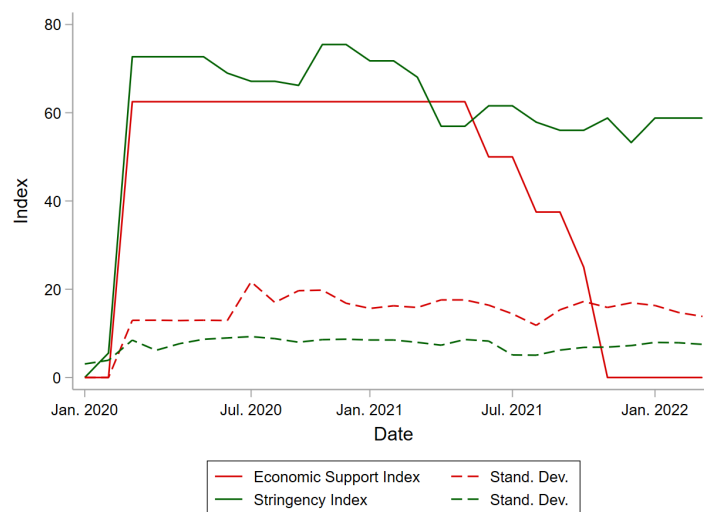


Notes: Time series of employment shares in selected industries for states grouped according to whether their 2017 age-adjusted opioid mortality is at or above the median or below it. Source: CPS.

A.4 Stringency Index and Economic Support Index

Figure A6 shows the average and standard deviation of the Stringency and Economic Support index developed by the Oxford Tracker. The Oxford Tracker Dataset categorizes various COVID-19 policies into indices representing different policy strengths, normalized between 0 and 100.

Figure A6: Stringency and Economic Support Indices



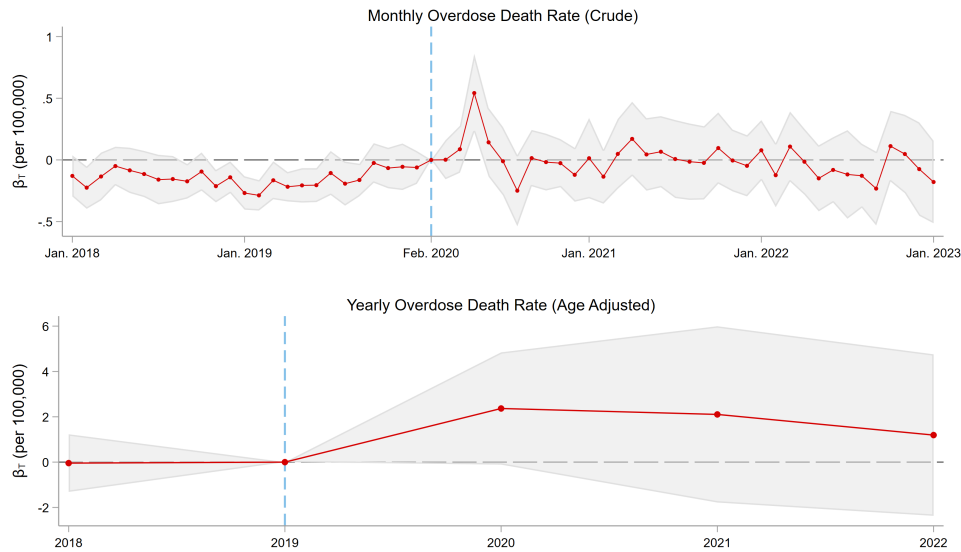
Notes: Summary policy indices for Stringency and Economic Support reported by the Oxford Tracker for the entire United States and the standard deviation of these policy indices across states. Source: Oxford Tracker.

B Placebos, Heterogeneity, and Robustness

B.1 Opioid Exposure and Deaths

Figure B1 shows the results obtained by estimating equation (1) using the overdose death rate as the dependent variable. The top panel uses the monthly overdose death rate, and the bottom panel uses the yearly age-adjusted overdose death rate. Initial opioid exposure is measured as a binary variable, with states being given a value of 1 if their age-adjusted opioid death rate in 2017 was at or above the median and 0 if below.

Figure B1: Non-Parametric Binary Event Study—Drug Overdose Deaths

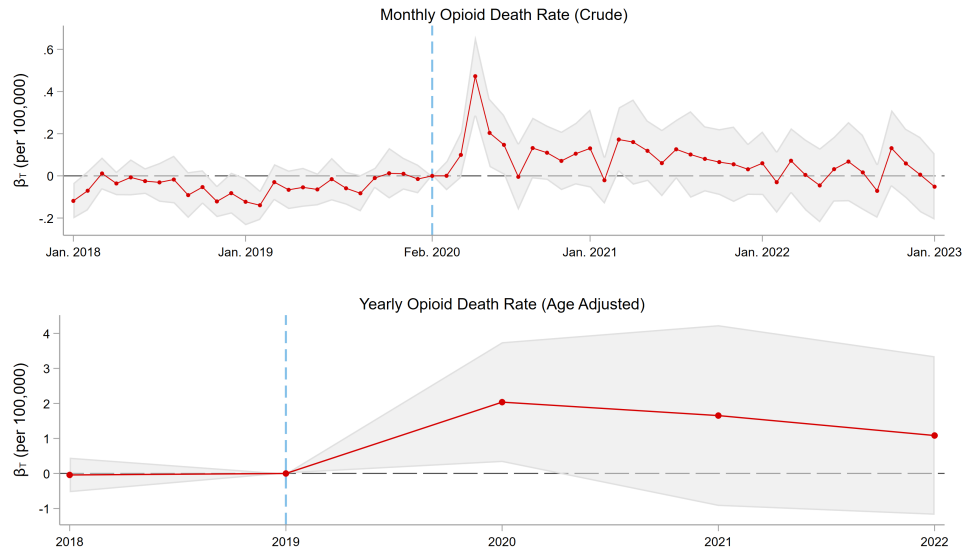


Notes: *Non-parametric event-study coefficients β_T and 95% confidence intervals. Coefficients represent changes in drug overdose death rates between states with different levels of prior opioid exposure relative to the relevant reference period. The top panel reports the monthly crude drug overdose death rate, for which February 2020 is the omitted reference month. The bottom panel reports the annual age-adjusted drug overdose death rate, for which 2019 is the omitted reference year. In both panels, coefficients are measured in deaths per 100,000 residents. The initial opioid exposure measure has been dichotomized: states with an age-adjusted opioid death rate in 2017 at or above the median are given a value of 1, the others a value of 0.*

Figure B2 shows the results obtained by estimating equation (1) using the opioid death rate as the dependent variable. The top panel uses the monthly opioid death rate, and the

bottom panel uses the yearly age-adjusted opioid death rate.

Figure B2: Non-Parametric Event Study—Opioid Overdose Deaths

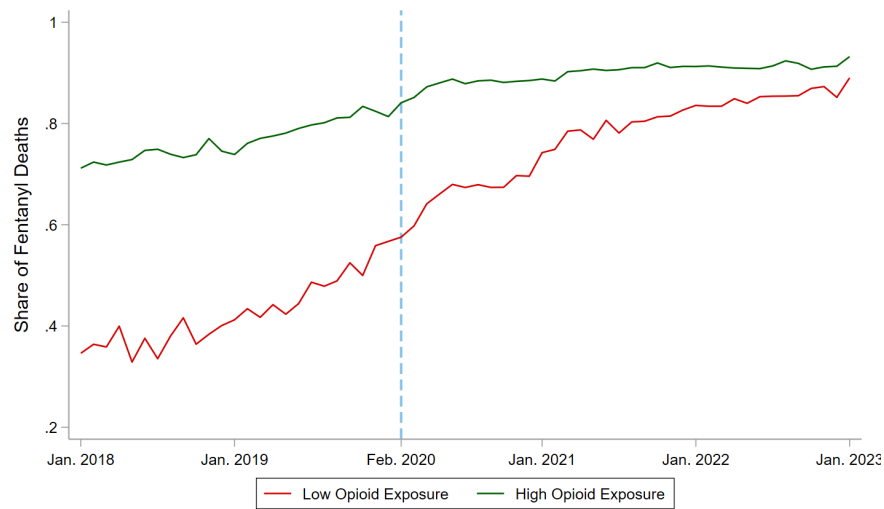


Notes: *Non-parametric event-study coefficients β_T and 95% confidence intervals. Coefficients represent changes in opioid overdose death rates between states with different levels of prior opioid exposure relative to the relevant reference period. The top panel reports the monthly crude opioid overdose death rate, for which February 2020 is the omitted reference month. The bottom panel reports the annual age-adjusted opioid overdose death rate, for which 2019 is the omitted reference year. In both panels, coefficients are measured in deaths per 100,000 residents. The initial opioid-exposure measure has been normalized to have a standard deviation equal to 1.*

Figure B3 reports the monthly share of opioid overdose deaths involving synthetic opioids (fentanyl) for the same high- and low-exposure groups. The fentanyl share is computed as the number of deaths involving synthetic opioids (ICD-10 code T40.4) divided by total opioid-related overdose deaths. This figure highlights how the composition of opioid deaths evolved across the two groups during the pandemic.

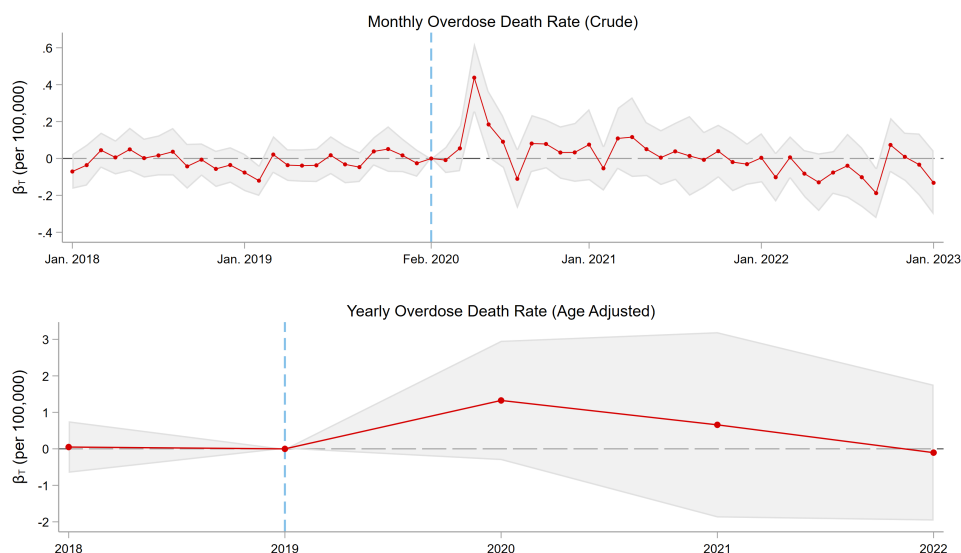
Figure B4 and Figure B5 show the results obtained by estimating equation (1) using the overdose death rate, and the opioid death rate as the dependent variable, respectively, but without including fentanyl-share as a control. The top panel uses the monthly overdose death rate, and the bottom panel uses the yearly age-adjusted overdose death rate.

Figure B3: Fentanyl Share by Opioid Exposure Group



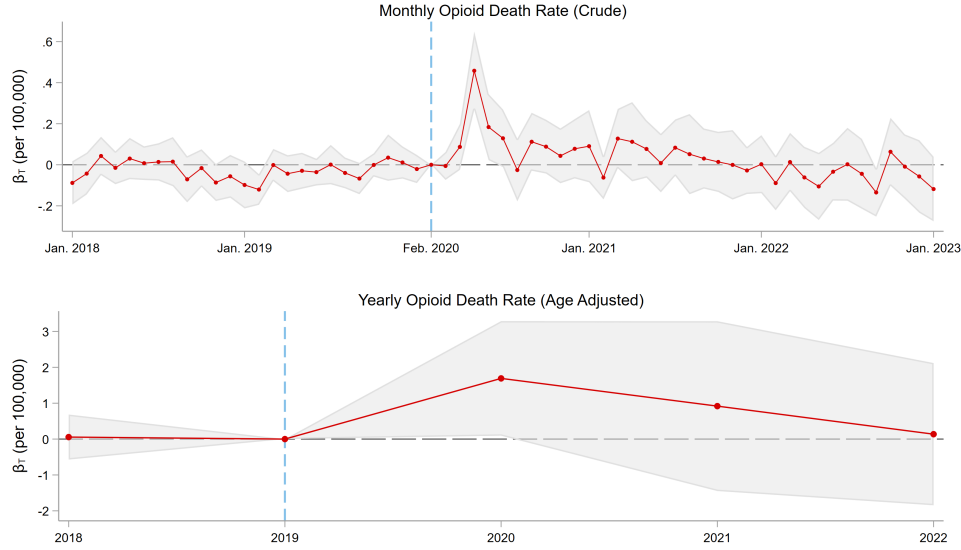
Notes: Monthly share of opioid overdose deaths involving synthetic opioids (fentanyl), computed as deaths with ICD-10 code T40.4 divided by total opioid-related overdose deaths. States are grouped into high and low exposure categories based on whether their 2017 age-adjusted opioid death rate is at or above the median or below it.

Figure B4: Non-Parametric Event Study—Drug Overdose Deaths, Without Fentanyl Control



Notes: Non-parametric event-study coefficients β_T and 95% confidence intervals. Coefficients represent changes in drug overdose death rates between states with different levels of prior opioid exposure relative to the relevant reference period. The top panel reports the monthly crude drug overdose death rate, for which February 2020 is the omitted reference month. The bottom panel reports the annual age-adjusted drug overdose death rate, for which 2019 is the omitted reference year. In both panels, coefficients are measured in deaths per 100,000 residents. The initial opioid-exposure measure has been normalized to have a standard deviation equal to 1.

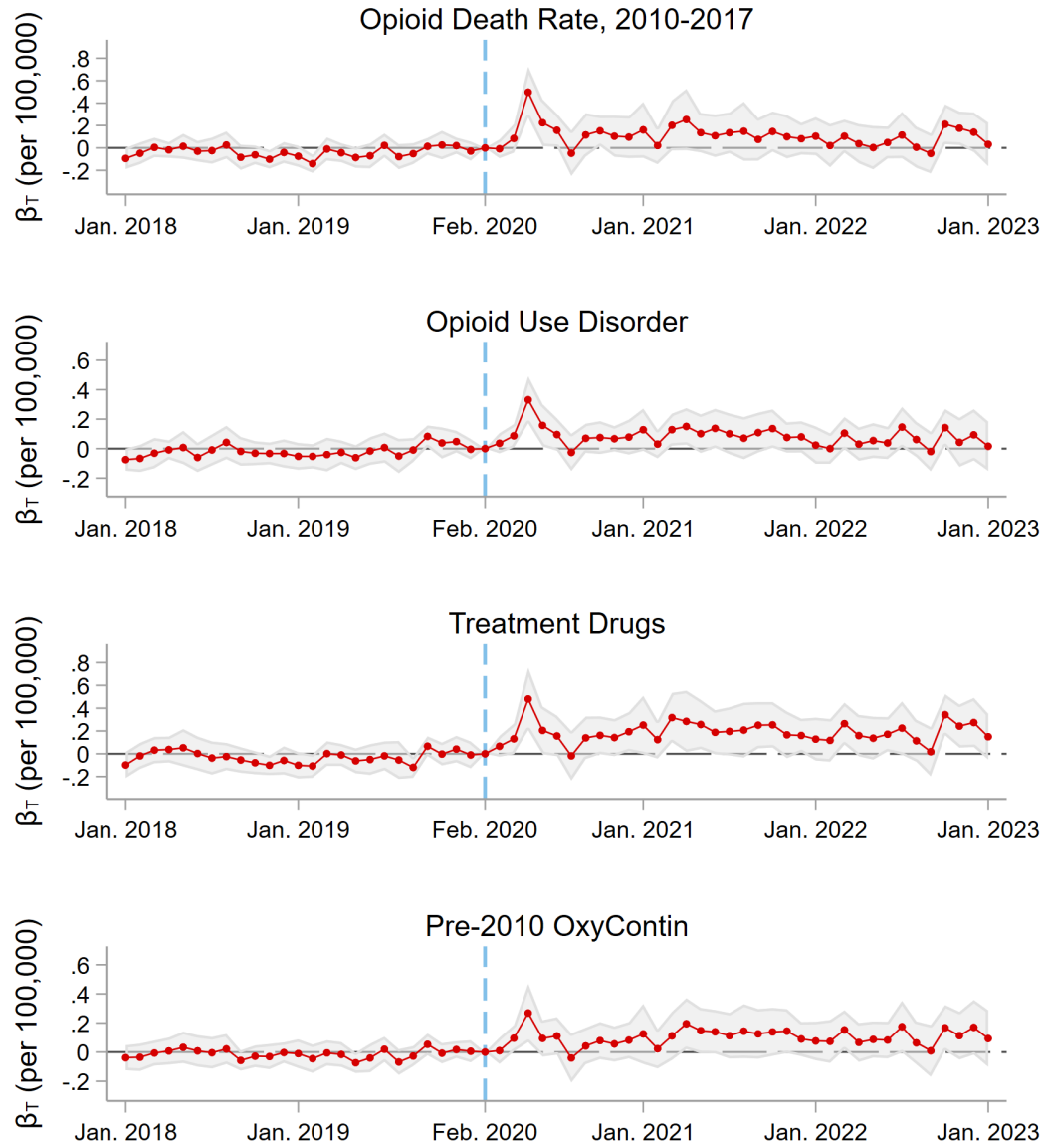
Figure B5: Non-Parametric Event Study—Opioid Overdose Deaths, Without Fentanyl Control



Notes: *Non-parametric event-study coefficients β_T and 95% confidence intervals. Coefficients represent changes in opioid overdose death rates between states with different levels of prior opioid exposure relative to the relevant reference period. The top panel reports the monthly crude opioid overdose death rate, for which February 2020 is the omitted reference month. The bottom panel reports the annual age-adjusted opioid overdose death rate, for which 2019 is the omitted reference year. In both panels, coefficients are measured in deaths per 100,000 residents. The initial opioid-exposure measure has been normalized to have a standard deviation equal to 1.*

Figure B6 shows the results obtained by estimating equation (1) using the monthly overdose death rate as the dependent variable and the four alternative measures of opioid exposure as independent variables. The top panel uses the combined opioid death rate between 2010 and 2017, the second panel the percentage of individuals with opioid use disorder in 2017-2018, the third panel uses the MME per capita of shipment of medications used to treat opioid addiction, and the bottom panel uses the pre-2010 OxyContin exposure.

Figure B6: Non-Parametric Event Study - Overdose Deaths with Alternative Measures

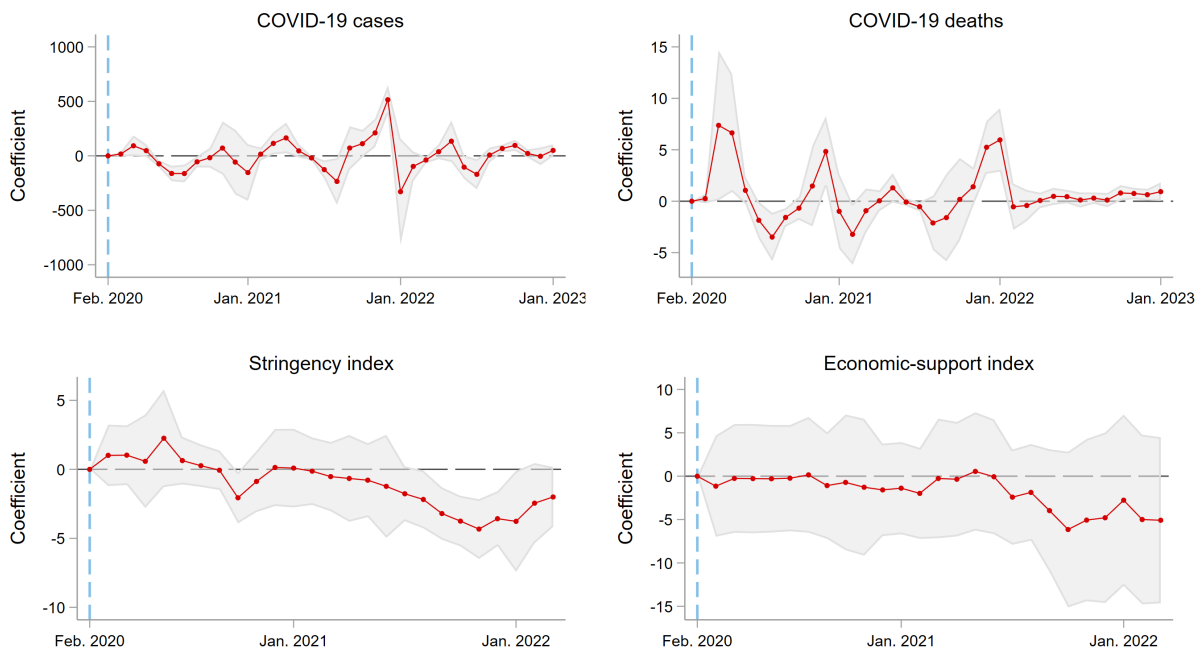


Notes: Non-parametric event study coefficients β_T and the 95% confidence intervals. Coefficients represent changes in overdose death rate relative to February 2020 between states with different levels of prior opioid exposure. Top panel uses opioid death for 2010-2017. Second panel uses percentage of people with opioid use disorder in 2017-2018. Third panel uses amount of MME per capita of drugs to treat opioid use disorder distributed to the state in 2017. The bottom panel uses pre-2010 OxyContin exposure. In all four cases, the measure has been normalized to have a standard deviation equal to 1.

B.2 Opioid Exposure and Pandemic Conditions

Figure B7 places monthly COVID-19 cases, monthly COVID-19 deaths, policy stringency, and economic support on the left-hand side of the non-parametric event study in equation (1). The estimated coefficients for cases and deaths are centered around zero, rather than exhibiting a consistently one-sided pattern. For the Stringency Index and the Economic Support Index, the coefficients are mostly statistically insignificant and tend to be negative. Thus, more exposed states do not appear to have adopted systematically more stringent containment policies or more generous economic support.

Figure B7: Opioid Exposure and Pandemic Conditions

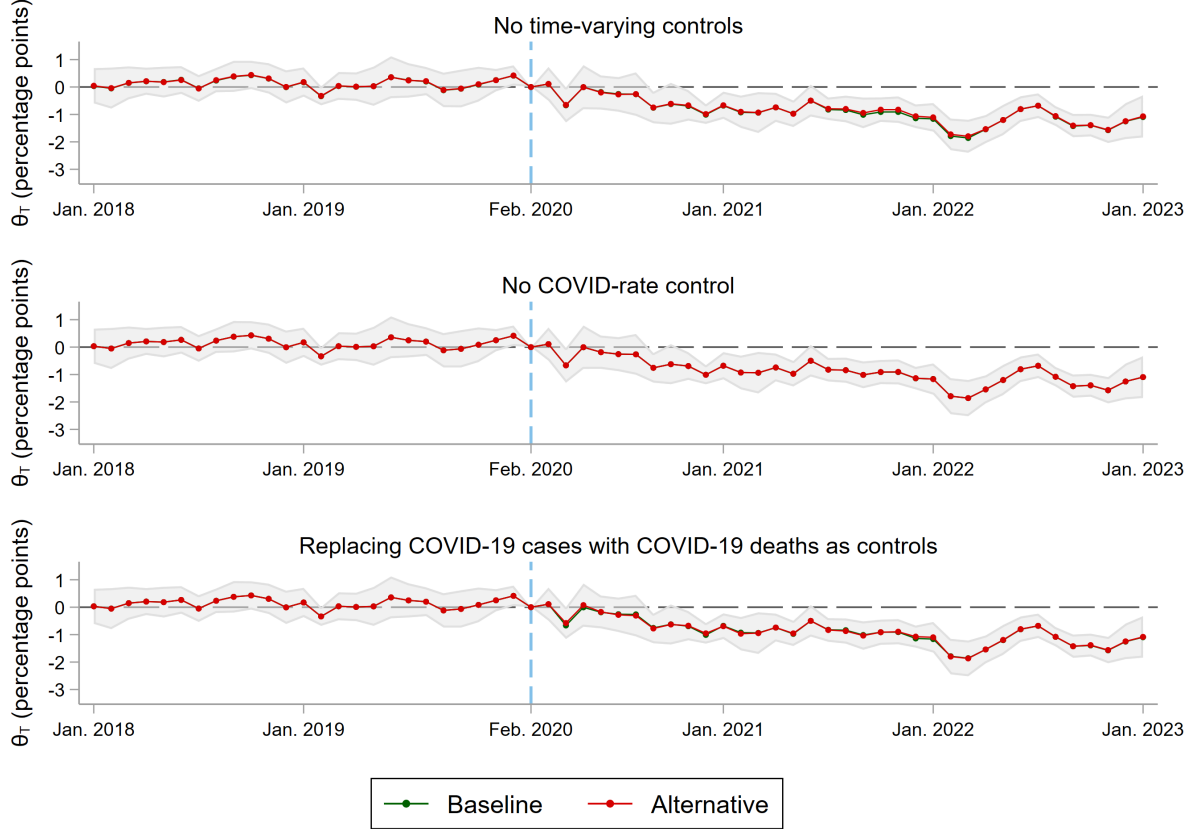


Notes: Non-parametric event-study coefficients β_T and state-clustered 95% confidence intervals. Coefficients represent changes in each pandemic outcome relative to February 2020 between states with different levels of prior opioid exposure. Cases and deaths are monthly flows per 100,000 residents. The exposure measure is the continuous 2017 age-adjusted opioid death rate, normalized to have a standard deviation equal to 1. No time-varying controls are included. The policy panels end in March 2022.

Figure B8 shows that the estimates are robust to alternative COVID-19-related controls. Replacing COVID-19 cases with COVID-19 deaths as a control for pandemic intensity leaves the estimated LFP decline essentially unchanged. The baseline full-post-period estimate is

−1.01 percentage points. It remains −1.01 percentage points when deaths replace cases or when the COVID-19 case control is omitted, and is −1.00 percentage points when all time-varying controls are omitted.

Figure B8: Main Estimates under Alternative Control Sets



Notes: Parametric event-study coefficients θ_T and state-clustered 95% confidence intervals. Coefficients represent average differences from state trends in prime-age LFP between states with different levels of prior opioid exposure. Each panel compares the baseline estimates in green with one alternative specification in red; grey bands show confidence intervals for the alternative. The alternatives omit all time-varying controls, omit only new monthly reported COVID-19 cases, or replace new monthly cases with new monthly COVID-19 deaths. The exposure measure is the continuous 2017 age-adjusted opioid death rate, normalized to have a standard deviation equal to 1.

B.3 Placebo Analysis

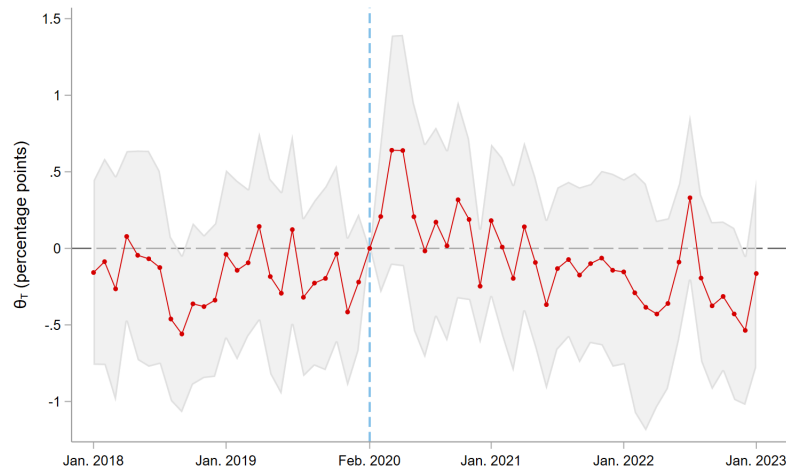
Table B1 shows the age-adjusted death rate in 2017 across the 15 leading causes of death in the US. The initial exposure measure used for the placebo analysis in the main text is constructed by aggregating the state-level death rate from these 15 causes, excluding the third (accidents) and eighth (intentional self-harm) causes of death. The reason for this is that some of the deaths in these causes are also included in the opioid death rate measure. Figure B9 presents the estimated coefficients from the parametric event study where the initial exposure measure is based on the total death rate from only the top 10 causes, excluding the third and eighth. Finally, Figure B10 shows the estimated coefficients from parametric event studies that have as the initial exposure measure the death rate of the cause highlighted in the title.

Table B1: Average Age-Adjusted Death Rate by Cause in 2017

	Death Rate
1. Diseases of heart (I00-I09,I11,I13,I20-I51)	165.99
2. Malignant neoplasms (C00-C97)	155.00
3. Accidents (unintentional injuries) (V01-X59,Y85-Y86)	54.01
4. Chronic lower respiratory diseases (J40-J47)	43.78
5. Cerebrovascular diseases (I60-I69)	37.43
6. Alzheimer disease (G30)	32.07
7. Diabetes mellitus (E10-E14)	21.99
8. Intentional self-harm (suicide) (*U03,X60-X84,Y87.0)	16.50
9. Influenza and pneumonia (J09-J18)	14.90
10. Nephritis (N00-N07,N17-N19,N25-N27)	12.79
11. Chronic liver disease and cirrhosis (K70,K73-K74)	11.30
12. Septicemia (A40-A41)	10.27
13. Parkinson disease (G20-G21)	8.62
14. Essential hypertension and hypertensive renal disease (I10,I12,I15)	8.44
15. Assault (homicide) (*U01-*U02,X85-Y09,Y87.1)	6.69

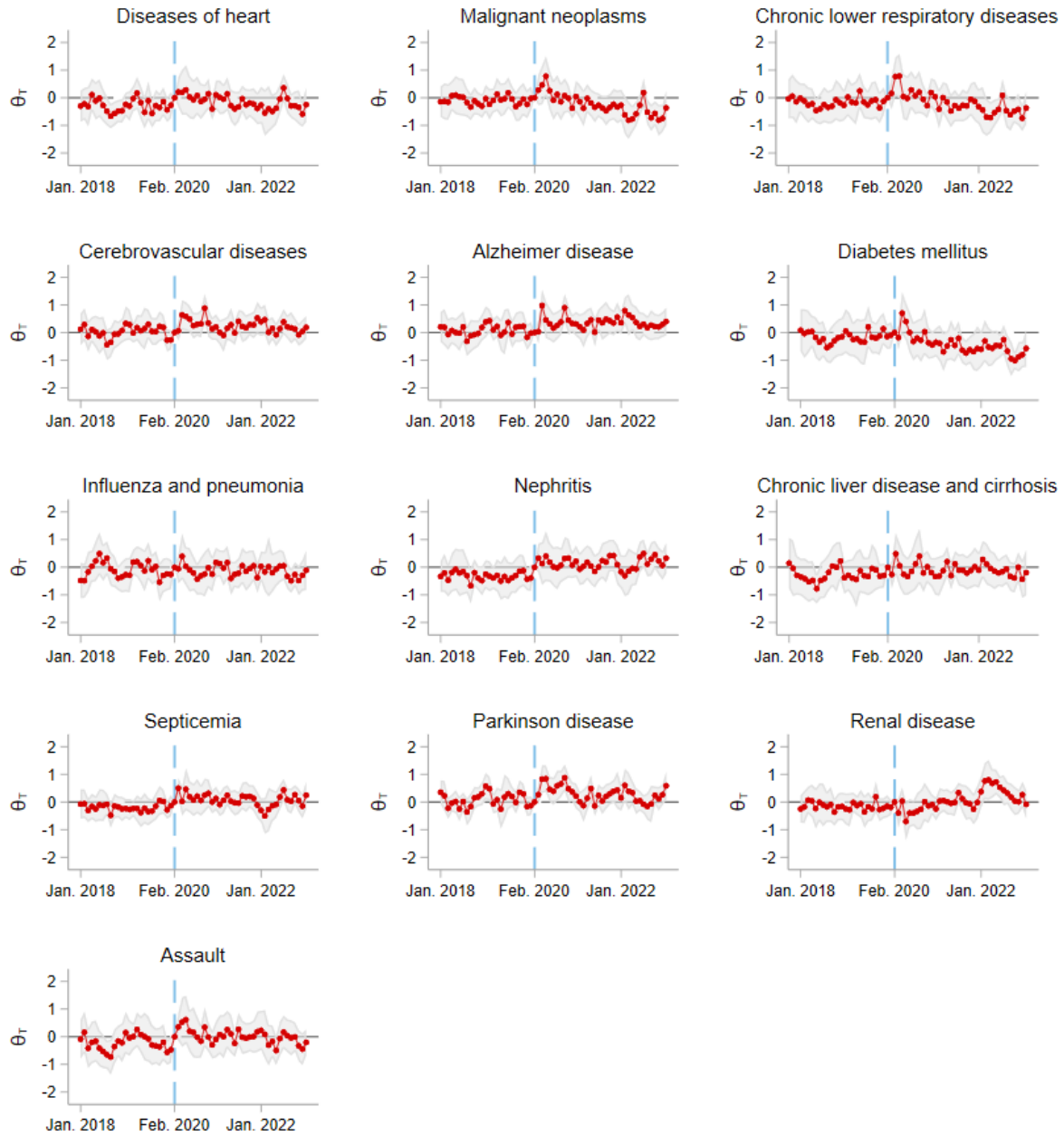
Notes: *The table shows the age-adjusted death rate in 2017 by the leading causes of death in the US. Source: CDC - Underlying Cause of Death.*

Figure B9: Parametric Event Study with Placebo Exposure



Notes: Placebo parametric event study coefficients θ_T and the 95% confidence intervals when using prime-age LFP as the dependent variable. Coefficients represent average differences from state-trend in prime-age LFP between states with different levels of prior placebo exposure. The measure of prior placebo exposure is the total age-adjusted death rate in 2017 due to the top 10 underlying causes of death, after excluding accidents and intentional self-harm. The death rate has been normalized to have a standard deviation equal to 1.

Figure B10: Parametric Event Study Estimates - Placebo

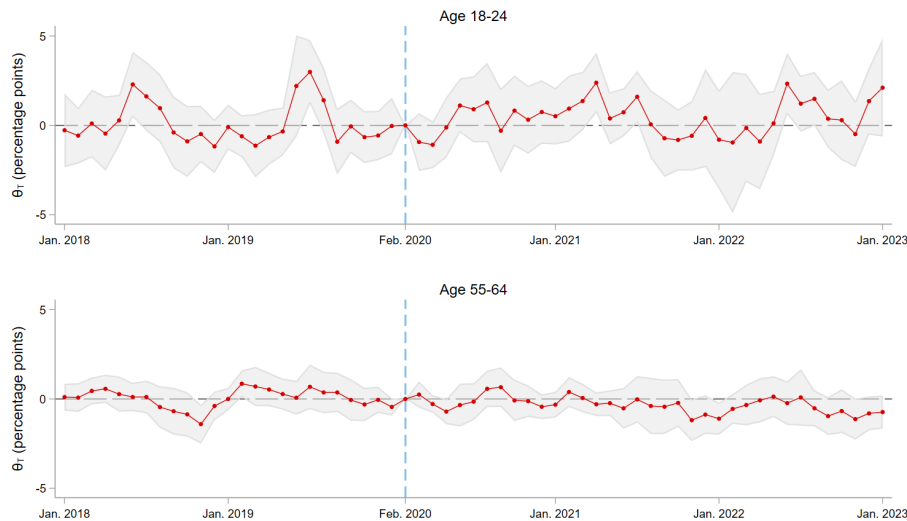


Notes: Placebo parametric event study coefficients θ_T and the 95% confidence intervals. Coefficients represent average differences from the state-trend in prime-age LFP between states with different levels of prior placebo exposure. The measure of prior placebo exposure is the age-adjusted death rate in 2017 due to causes indicated in the title of each panel. All death rates have been normalized to have a standard deviation equal to 1.

B.4 Alternative Age Groups

The main analysis focuses on the labor-force participation of prime-age workers, i.e., ages between 25 and 54. Figure B11 shows the estimated coefficients from the parametric event study analysis when focusing on the age groups 18-24 and 55-64.

Figure B11: Parametric Event Study Estimates - Labor Force Across Samples

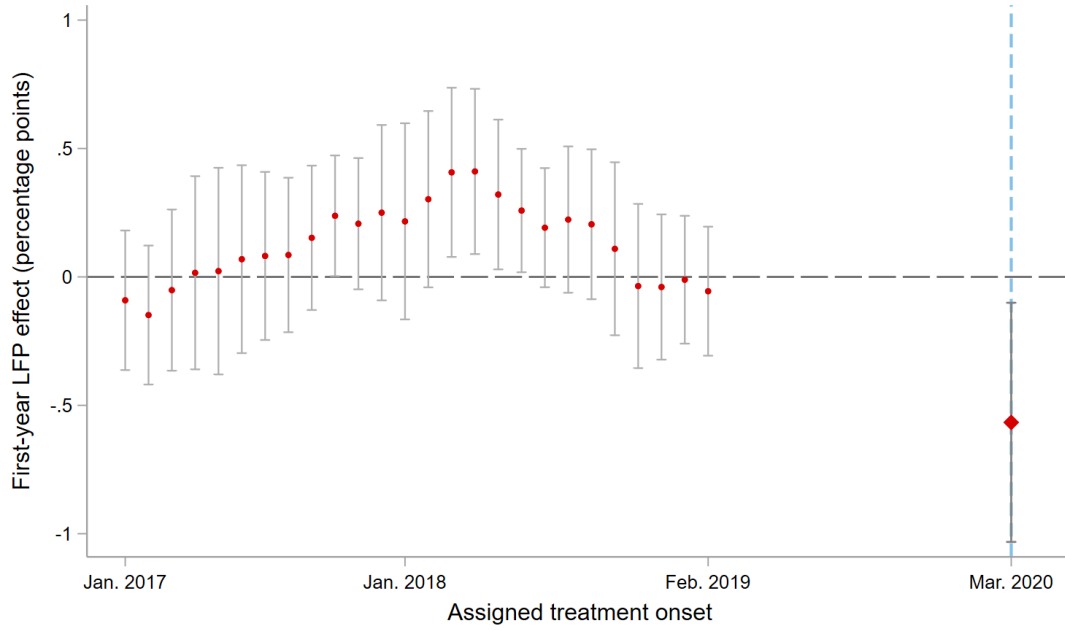


Notes: Parametric event study coefficients θ_T and the 95% confidence intervals. Coefficients represent average differences from state-trend in LFP between states with different levels of prior opioid exposure. The upper panel restricts the sample to individuals aged 18-24, and the lower panel to those 55-64. The initial opioid exposure measure has been normalized to have a standard deviation equal to 1.

B.5 Placebo Treatment Dates

Figure B12 presents the results from estimating the parametric event study for each of the 26 placebo treatment dates. Each estimate treats one month from January 2017 through February 2019 as the onset of treatment, uses the preceding 24 months as the pre-period, and averages the estimated effect over the assigned month and the following 11 months. None of these pre-COVID placebo dates yields a statistically significant negative estimate. The estimate obtained by applying the identical 12-month specification to the actual March 2020 onset is shown for comparison.

Figure B12: Placebo Treatment Dates

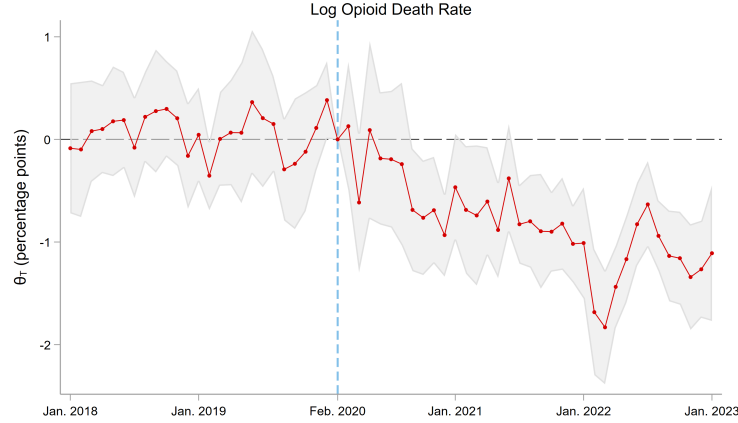


Notes: *Parametric event-study estimates and state-clustered 95% confidence intervals. Each point reports the average LFP effect during the first 12 months beginning with the assigned onset. The exposure measure is the continuous 2017 age-adjusted opioid death rate, normalized to have a standard deviation equal to 1. All specifications include state and month fixed effects, state-specific linear trends, population weights, and the employment-based Bartik control. Realized COVID-19 and pandemic-policy controls are omitted so that the placebo and March 2020 windows are comparable. The light-blue line identifies the actual March 2020 onset.*

B.6 Logarithmic Opioid-Exposure Measure

Figure B13 shows the estimated coefficients from the parametric event study analysis when using the logarithm of the age-adjusted opioid death rate in 2017.

Figure B13: Parametric Event Study Estimates - Log Opioid-Overdose Death Rate

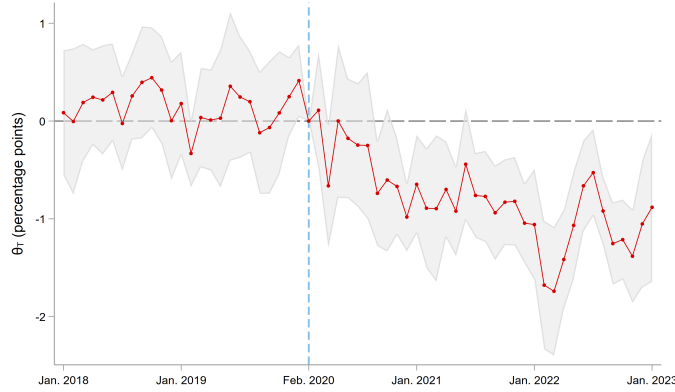


Notes: *Parametric event study coefficients θ_T and the 95% confidence intervals. Coefficients represent average differences from state-trend in prime-age LFP between states with different levels of prior opioid exposure, measured by the log of the 2017 opioid-related overdose age-adjusted death rate, normalized to have a standard deviation equal to 1.*

B.7 Alternative Specifications

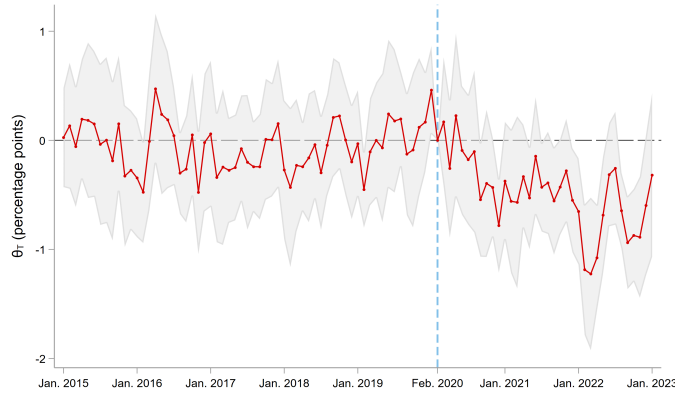
Figure B14 shows the estimated coefficients from estimating the parametric event study analysis, allowing for quadratic rather than linear state-specific trends. Figure B15 shows the estimated coefficients from estimating the parametric event study using a longer pre-period. Instead of estimating trends over 2018–2023 and using the 2017 opioid death rate as the exposure measure, the specification is re-estimated using data beginning in 2015 and using the 2015 opioid death rate as the measure of pre-COVID exposure. Under these alternative specifications, the results remain consistent with the main findings.

Figure B14: Parametric Event Study Estimates – Quadratic State Trends



Notes: *Parametric event study coefficients θ_T and 95% confidence intervals estimated from a specification that allows for quadratic state-specific trends. Coefficients represent average differences from state trends in prime-age LFP between states with different levels of prior opioid exposure. The initial opioid exposure measure has been normalized to have a standard deviation of 1.*

Figure B15: Parametric Event Study Estimates – Longer Pre-Period (2015 Start)



Notes: *Parametric event study coefficients θ_T and 95% confidence intervals estimated using data from 2015–2023 and the 2015 opioid-related overdose age-adjusted death rate as the exposure measure. Coefficients represent average differences from state trends in prime-age LFP between states with different levels of prior opioid exposure. Exposure is normalized to have a standard deviation of 1.*

C Data on Mechanisms

C.1 Employment and Health by Drug-Use Status

Tables C1 and C2 present data on employment, health, and work disability for adults aged 24–49, based on the 2019 and 2021/2022 waves of the NSDUH. Individuals are categorized by their drug use in the past 12 months. The nonusers group includes those who reported no illicit drug use, excluding marijuana, in the past 12 months; prescription drugs count only when misused. The disorder group comprises individuals diagnosed with a drug use disorder, excluding marijuana use disorder.

Table C1: Employment and Health Status by Drug Use, Ages 24-49

<i>Panel A</i>		2019		2021/2022		
	Emp.	Unemp.	Out of LF	Emp.	Unemp.	Out of LF
Nonuser	79.3%	3.9%	16.8%	75.8%	5.1%	19.1%
Disorder	58.7%	15.3%	26.0%	54.3%	13.0%	32.7%
Population	79.0%	4.4%	16.6%	75.0%	5.7%	19.3%

<i>Panel B</i>		2019		2021/2022		
	Very Good	Good	Fair/Poor	Very Good	Good	Fair/Poor
Nonuser	61.4%	28.7%	9.9%	58.1%	31.2%	10.7%
Disorder	39.5%	35.0%	25.5%	33.2%	37.4%	29.4%
Population	60.5%	28.9%	10.6%	56.5%	31.7%	11.8%

Notes: The table shows the employment status in Panel A, and self-reported health status in Panel B, by type of drug user in 2019 and for the 2021/2022 period. Self-reported health is grouped into three categories: “Very Good” (including excellent and very good health), “Good,” and “Fair/Poor” (including fair and poor health). The type of drug user is defined by the NSDUH. See the text for further details. Nonusers did not use any drug (excluding marijuana) in the past 12 months. For each drug user type and year, the row sums up to 100. Source: NSDUH.

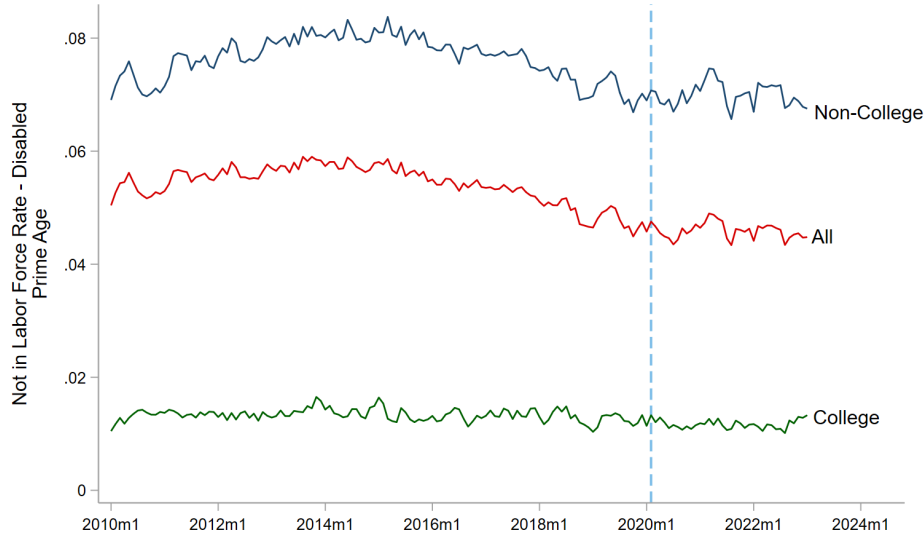
C.2 Disability-Related Labor-Force Nonparticipation

Figure C1 shows the share of prime-age individuals who are out of the labor force due to disability.

Table C2: Drug Use and Work Disability, Ages 24-49

	NLF - Disability		Work Days Sick		Work Days Skip	
	2019	2021/2022	2019	2021/2022	2019	2021/2022
Nonuser	17.4%	16.8%	0.83	1.00	0.28	0.32
Disorder	25.1%	44.1%	2.12	2.08	1.94	0.96
Population	18.2%	19.4%	0.88	1.05	0.34	0.38

Notes: The table shows several measures of absence from work by type of drug user in 2019 and for the 2021/2022 period. The first two columns show, among individuals out of the labor force, the percentage who report disability as the reason for nonparticipation. The second two columns show the average number of work days missed in the past 30 days due to sickness. The last two columns show the average number of work days missed for other reasons. The type of drug user is defined by the NSDUH. See the text for further details. Source: NSDUH.

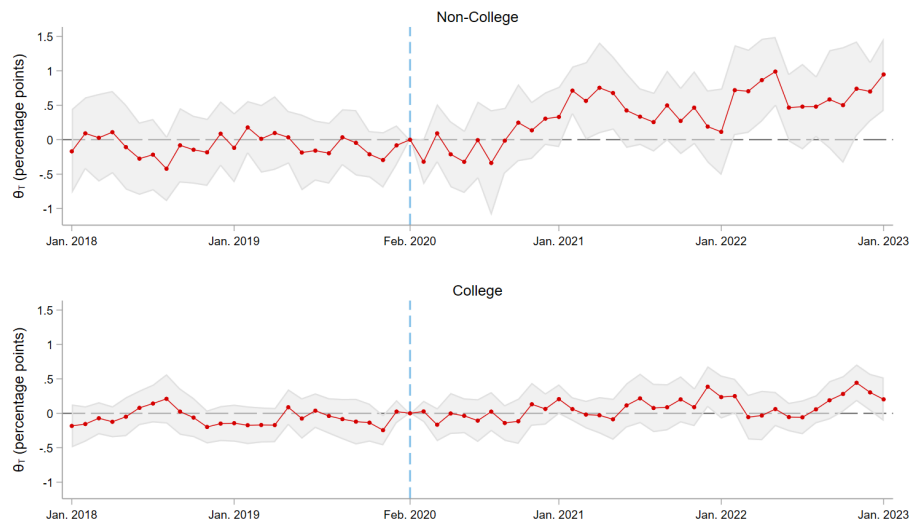
Figure C1: Out-of-labor-force due to Disability Rate

Notes: Share of prime-age individuals not in the labor force for disability reasons. This measure includes individuals who report being unable to work or disability as their reason for nonparticipation. Source: CPS

C.3 Disability-Related Labor-Force Nonparticipation by Education

Figure C2 shows the estimated coefficients from conducting separate parametric event studies for the non-college and college-educated samples using the fraction out-of-labor-force due to disability as the dependent variable.

Figure C2: Parametric Event Study Estimates - Out-of-LF due to Disability



Notes: *Parametric event-study coefficients θ_T and the 95% confidence intervals. Coefficients represent average differences from state-trend in prime-age out of labor force due to disability between states with different levels of prior opioid exposure. The top panel uses prime-age out of labor force due to disability among non-college-educated, while the bottom uses college-educated. The initial opioid exposure measure has been normalized to have a standard deviation equal to 1.*